On Merca's stronger conjecture for the truncated Jacobi triple product series

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Abstract. In the study of theta series and partition functions, Andrews and Merca as well as Guo and Zeng independently conjectured that a suitable truncation of the Jacobi triple product series has nonnegative coefficients. In 2015 this conjecture was proved analytically and combinatorially by Mao and Yee, respectively. In 2021, Merca proposed a stronger version of the conjecture, stating that for positive integers R, S, k such that $S < R$, the coefficient of q^n in the theta series

$$\frac{(-1)^k \sum_{j=k}^{\infty}(-1)^j q^{Rj(j+1)/2}\left(q^{-Sj}-q^{(j+1)S}\right)}{(q^S, q^{R-S}; q^R)_\infty}$$

is nonnegative. Recently, some special cases of this conjecture were proved by Ballantine and Feigon. In this paper, based on the partition theoretical method and the circle method, we show that for $1 \le S \le R/2$, Merca's stronger conjecture holds provided $n \ge N$, where N is an explicitly determined integer, depending on R, S and k.

Keywords: truncated theta series; partition function; Jacobi triple product series; nonmodular infinite product; circle method

AMS Classification: 05A17, 33D15, 11P55, 11P82

1 Introduction

In 2012, Andrews and Merca investigated a truncated form of Euler's pentagonal number theorem and obtained the following result [2, Lemma 1.2]:

$$\frac{(-1)^{k-1}}{(q;q)_\infty}\sum_{j=-k}^{k-1}(-1)^j q^{j(3j+1)/2} = (-1)^{k-1} + \sum_{n=1}^{\infty}\frac{q^{\binom{k}{2}+(k+1)n}}{(q;q)_n}\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q. \tag{1.1}$$

The truncated sum on the left hand side can be written in the form of

$$\sum_{j=-k-\tau}^{k}(-1)^j q^{aj^2+bj}$$

by replacing k with $k+1$, where $a \in \mathbb{Q}^+, b \in \mathbb{Q}$ and $\tau \in \{0,1\}$. Such truncated sums can be traced back to Shanks' version [31, (2)] of Euler's pentagonal number theorem,

$$\sum_{j=-k}^{k}(-1)^j q^{j(3j+1)/2} = \sum_{n=0}^{k}\frac{(-1)^n(q;q)_k q^{\binom{n}{2}+(k+1)n}}{(q;q)_n}.$$

Warnaar also studied the case of $\sum_{j=-k-\tau}^{k}(-1)^j q^{j(5j+c)/2}$ in [33, Theorem 1.1] with $c \in \{1,3\}$, for which Chapman [9] provided a combinatorial proof.

One can see that the coefficient of q^n on the right hand side of (1.1) is nonnegative for $n \ge 1$, for which Andrews and Merca also gave a corresponding partition interpretation, which also leads to

$$(-1)^{k-1}\sum_{j=0}^{k-1}(-1)^j\big(p(n-j(3j+1)/2) - p(n-j(3j+5)/2-1)\big) \ge 0.$$

Later, Guo and Zeng studied a class of truncated forms of Gauss' triangular and square numbers series in [16], from which they also derive the recurrence relations for partition functions. Since then the truncated forms for theta series have attracted more and more attention.

In 2018, Andrews and Merca [3] obtained the unified truncated forms for Euler's pentagonal number theorem, Gauss's theta series for triangular and square numbers, which are further generalized by Xia, Yee and Zhao in 2022 [34]. We can easily notice that these three classical theta series all arise from the Jacobi triple product series. Actually, Andrews and Merca [2], and Guo and Zeng [16] proposed the following conjecture for a truncated Jacobi triple product series, independently.

Conjecture 1.1. *For positive integers k, R, S with $1 \leq S < R/2$, the coefficient of q^n with $n \geq 1$ in*

$$(-1)^{k-1} \frac{\sum_{j=0}^{k-1} (-1)^j q^{Rj(j+1)/2-Sj} \left(1-q^{(2j+1)S}\right)}{\left(q^S, q^{R-S}, q^R; q^R\right)_\infty} \tag{1.2}$$

is nonnegative.

The conjecture was proved analytically by Mao [22] and combinatorially by Yee [36] in 2015. By applying the theory of Bailey pairs and Liu's expansion formula [20] for q-series, Wang and Yee [32] reproved this conjecture. Recently, by utilizing the iteration of Bailey's methods, we [12] obtained a general expression of (1.2) in the form of multiple sums and the results can reduce to Andrews-Gordon and Bressoud type identities.

In 2021, Merca [24, 25] proposed a stronger version of Conjecture 1.1 as follows.

Conjecture 1.2. *For positive integers $1 \leq S < R$ with $k \geq 1$, the theta series*

$$\mathcal{H}_{R,S,k}(q) := (-1)^k \frac{\sum_{j=k}^{\infty} (-1)^j q^{Rj(j+1)/2} \left(q^{-Sj} - q^{(j+1)S}\right)}{\left(q^S, q^{R-S}; q^R\right)_\infty} \tag{1.3}$$

has nonnegative coefficients.

Ballantine and Feigon [6] proved the conjecture combinatorially for the cases when $k \in \{1, 2, 3\}$. We also confirmed the cases when $R = 3S$ for $S \geq 1$ in [12] by using the partition theoretical method. For the related works, see also Yao [35], Ballantine and Feigon [6], and Zhou [37]. Mao subsequently proved eventual nonnegativity by Wright's circle method [23]. On the one hand, we analyse the generating functions for restricted partition functions; on the other hand, asymptotic formulae for nonmodular infinite products are given.

Although replacing S by $R-S$ interchanges the two residue classes in the denominator, it does not leave the truncated numerator unchanged. We therefore give the proof for $1 \leq S < R/2$. If $d = (R, S)$, $R = dr$, and $S = ds$, then

$$\mathcal{H}_{R,S,k}(q) = \mathcal{H}_{r,s,k}(q^d).$$

Thus it is enough in the proof to assume $(r, s) = 1$. The boundary case $R = 2S$ reduces to $r = 2, s = 1$ and is handled separately in Corollary 4.5.

Conjecture 1.3. *For positive integers $1 \leq s < r/2$, $(r, s) = 1$ and $k \geq 1$, the coefficient of q^n in*

$$(-1)^k \frac{\sum_{j=k}^{\infty} (-1)^j q^{rj(j+1)/2} \left(q^{-sj} - q^{(j+1)s}\right)}{\left(q^s, q^{r-s}; q^r\right)_\infty} \tag{1.4}$$

is nonnegative.

In this paper, we seek to determine a nonnegative integer $N(r, s, k)$ depending on r, s, k such that for such given r, s and k, Merca's stronger conjecture holds for $n \geq N(r, s, k)$.

Theorem 1.4. *For positive integers R, S, k with $1 \le S \le R/2$, there is an explicit integer $N(R,S,k) \ge 0$ such that, whenever $n \ge N(R,S,k)$, the coefficient of q^n in*

$$(-1)^k \frac{\sum_{j=k}^{\infty} (-1)^j q^{Rj(j+1)/2} \left(q^{-Sj} - q^{(j+1)S}\right)}{(q^S, q^{R-S}; q^R)_\infty} \tag{1.5}$$

is nonnegative.

To do so, we first decompose the denominator of the above theta series into two factors as follows:

$$\frac{1}{(q^s;q^r)_\alpha (q^{r-s};q^r)_\beta} \frac{1}{(q^{\alpha r+s}, q^{(\beta+1)r-s}; q^r)_\infty}, \tag{1.6}$$

where $(\alpha,\beta) = (2,2)$ in the pairwise-coprime four-part case, $(\alpha,\beta) = (2,1)$ in the remaining case with $2 \mid s$, and $(\alpha,\beta) = (1,2)$ in the remaining case with $2 \nmid s$. The first factor is treated by restricted partitions. Its coefficient function is a polynomial plus an exactly periodic error and the extrema over one period give the upper and lower bounds used in the four cases of Sections 3 and 4. The second factor in (1.6) is a nonmodular infinite product.

The asymptotic study of nonmodular infinite products goes back to Lehner [19], Livingood [21], and Grosswald [15]. More recently, Chern [10] used this method in work on a conjecture of Seo and Yee [30]; Craig later settled that conjecture completely [11].

For positive integers $1 \le a < M/2$ with $(a,M) = 1$, denote

$$G_{a,M}(q) := \frac{1}{(q^{2M+a}, q^{3M-a}; q^M)_\infty} = \sum_{n=0}^{\infty} g_{a,M}(n) q^n, \tag{1.7}$$

and write

$$\Phi_{a,M}(q) = \log \frac{1}{(q^a;q^M)_\infty}, \qquad \Psi_{a,M}(q) = \log \frac{1}{(q^{2M+a};q^M)_\infty}. \tag{1.8}$$

We obtain the following asymptotic formula for the coefficient of $G_{a,M}(q)$.

Theorem 1.5. *Let $M \ge 4$, $1 \le a < M/2$, and $(a,M) = 1$. When n is sufficiently large,*

$$g_{a,M}(n) \sim \frac{\Delta_{a,M}}{2\sin(a\pi/M)} \left(\frac{\pi^2}{3Mn}\right)^{5/2} I_{-5}\left(2\pi\sqrt{\frac{n}{3M}}\right),$$

where I_ν is the modified Bessel function of the first kind and $\Delta_{a,M} := a(M-a)(M+a)(2M-a)$.

Replacing a, M by s, r, respectively, also gives effective two-sided bounds for $g_{s,r}(n)$.

Theorem 1.6. *Let $r \ge 4$, $1 \le s < r/2$, and $(r,s) = 1$. For $n \ge T_{\mathrm{s}}(r)$,*

$$g_{s,r}(n) \ge \frac{0.99\pi^4\Delta}{4(3r)^{9/4}\sin(s\pi/r)} n^{-11/4} \exp\left(2\pi\sqrt{\frac{n}{3r}}\right) =: g^d(n),$$
$$g_{s,r}(n) \le \frac{1.01\pi^4\Delta}{4(3r)^{9/4}\sin(s\pi/r)} n^{-11/4} \exp\left(2\pi\sqrt{\frac{n}{3r}}\right) =: g^u(n),$$

where $\Delta := s(r-s)(r+s)(2r-s)$ and $T_{\mathrm{s}}(r)$ is defined in Section 6.

Finally, we combine the finite estimates of Sections 3 and 4 with the coefficient bounds in Theorem 1.6 to prove Theorem 1.4.

This paper is organized as follows. Section 2 recalls the definitions and analytic tools used below. Sections 3 and 4 treat the finite restricted-partition factors through the four cases. Section 5 discusses the nonmodular infinite products, their bounds on the minor arcs, and their approximations near the dominant singularity. Section 6 derives the coefficient estimates of the nonmodular infinite products and proves Theorem 1.4. The supplementary Mathematica file `Merca_Original_Notebook_Updated.nb`, together with its English user guide, reproduces the necessary evaluations.

2 Preliminaries

Throughout this paper, we adopt standard notations and terminologies for q-series, see, for example, [14]. We assume that $|q| < 1$. The q-shifted factorial is denoted by

$$(a)_n = (a;q)_n = \begin{cases} 1, & \text{if } n = 0, \\ (1-a)(1-aq)\cdots(1-aq^{n-1}), & \text{if } n \geq 1. \end{cases}$$

We also use the notation

$$(a)_\infty = (a;q)_\infty = \prod_{n=0}^{\infty}(1-aq^n).$$

There are more compact notations for the multiple q-shifted factorials

$$(a_1, a_2, \ldots, a_m; q)_n = (a_1;q)_n (a_2;q)_n \cdots (a_m;q)_n,$$
$$(a_1, a_2, \ldots, a_m; q)_\infty = (a_1;q)_\infty (a_2;q)_\infty \cdots (a_m;q)_\infty.$$

The well-known Jacobi triple product identity is given by [14, II.28]

$$\sum_{k=-\infty}^{\infty} (-1)^k q^{\binom{k}{2}} a^k = (a, q/a, q; q)_\infty. \tag{2.1}$$

Truncated theta series are closely related to integer partitions. Recall that a partition of a nonnegative integer n is a finite nonincreasing sequence of positive integers $\lambda_1, \lambda_2, \ldots, \lambda_k$ such that $\sum_{i=1}^{k} \lambda_i = n$, see, for example, [1]. Let $p(n)$ denote the number of integer partitions of n with $p(0) = 1$ and $p(n) = 0$ for $n < 0$, which has the generating function

$$\sum_{n=0}^{\infty} p(n) q^n = \frac{1}{(q;q)_\infty}.$$

We then reveal the following important result given by Pólya and Szegő.

Lemma 2.1 ([28, Part I, Problem 27.1])**.** *Let $a_1, a_2, \ldots, a_\ell$ be pairwise-coprime positive integers. Then*

$$\frac{1}{(1-q^{a_1})(1-q^{a_2})\cdots(1-q^{a_\ell})} = R\left(\frac{1}{1-q}\right) + \frac{S(q)}{1-q^{a_1 a_2 \cdots a_\ell}}, \tag{2.2}$$

where R is a polynomial of degree ℓ, while S is a polynomial of degree less than $a_1 a_2 \cdots a_\ell$.

Let $p_\ell(n)$ count the solutions of $a_1 x_1 + \cdots + a_\ell x_\ell = n$ in nonnegative integers $x_1, \ldots, x_\ell$. For distinct a_i, this counts partitions with parts in $\{a_1, \ldots, a_\ell\}$. Its generating function is

$$\sum_{n=0}^{\infty} p_\ell(n) q^n = \frac{1}{(1-q^{a_1})(1-q^{a_2})\cdots(1-q^{a_\ell})}. \tag{2.3}$$

Lemma 2.2. *Let $\ell \geq 2$, let $a_1, \ldots, a_\ell$ be pairwise-coprime positive integers, and put $A = a_1 \cdots a_\ell$. Let $\mathscr{P}_{\ell-1}(n)$ be the coefficient of q^n in the full principal part of $\prod_{i=1}^{\ell}(1-q^{a_i})^{-1}$ at $q = 1$. Then*

$$p_\ell(n) = \mathscr{P}_{\ell-1}(n) + \mathscr{E}_{\boldsymbol{a}}(n), \qquad \mathscr{E}_{\boldsymbol{a}}(n) = \sum_{\substack{1 \leq i \leq \ell \\ a_i > 1}} \frac{1}{a_i} \sum_{\substack{\zeta^{a_i} = 1 \\ \zeta \neq 1}} \frac{\zeta^{-n}}{\prod_{j \neq i}(1-\zeta^{a_j})}. \tag{2.4}$$

The function $\mathscr{E}_{\boldsymbol{a}}$ is A-periodic and satisfies

$$|\mathscr{E}_{\boldsymbol{a}}(n)| \leq \mathscr{B}_\ell(\boldsymbol{a}) := \frac{1}{4^{\ell-1}} \sum_{i=1}^{\ell} a_i^{\ell-2}(a_i - 1). \tag{2.5}$$

Proof. Pairwise coprimality makes every pole away from $q = 1$ simple. At a nontrivial a_ith root ζ,

$$\lim_{q\to\zeta} \frac{1-q/\zeta}{1-q^{a_i}} = \frac{1}{a_i}$$

and the principal part is therefore

$$\frac{1}{a_i \prod_{j\neq i}(1-\zeta^{a_j})} \frac{1}{1-q/\zeta},$$

whose coefficient of q^n is the summand in (2.4). This proves the formula and also its A-periodicity.

It remains to estimate the summands. Write $\zeta = e^{2\pi i m/a_i}$, where $1 \le m < a_i$. Since $(a_i, a_j) = 1$, the least positive residue of ma_j modulo a_i is nonzero, and hence

$$|1-\zeta^{a_j}| = 2\left|\sin\frac{\pi m a_j}{a_i}\right| \ge 2\sin\frac{\pi}{a_i} \ge \frac{4}{a_i}.$$

Here the last inequality follows from $\sin x \ge 2x/\pi$ for $0 \le x \le \pi/2$. Taking absolute values in (2.4), and noting that there are $a_i - 1$ nontrivial a_ith roots, gives (2.5). □

Next, we introduce the notation used below. When parentheses are used in greatest-common-divisor notation, $(a_1, \ldots, a_\ell)$ denotes the greatest common divisor of the entries, while $a \equiv b \pmod m$ has its usual meaning. We write $\Re z$ and $\Im(z)$ for the real and imaginary parts of z. The notations $F = O(G)$ and $F \ll G$ mean that $|F| \le C|G|$, where C is independent of the limiting variable. When C depends on y, we write $F = O_y(G)$ or $F \ll_y G$.

Recall that the Gamma function $\Gamma(x)$ [5] is given by

$$\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}dt,$$

where $\Re(x) > 0$. It satisfies the reflection formula [4, p. 250, eq. (3)]

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin(\pi x)}. \tag{2.6}$$

It is also known that $\Gamma(n) = (n-1)!$ for a positive integer $n \ge 1$ and so $\Gamma(1) = 1$.

If f is holomorphic on a neighbourhood of the closed disk bounded by a positively oriented circle C, then Cauchy's integral formula gives

$$f(z_0) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-z_0}\,dz \tag{2.7}$$

for every z_0 inside C. More generally, if f is meromorphic inside C, holomorphic on C, and has poles $z_1, \ldots, z_k$ in its interior, then the residue theorem gives

$$\frac{1}{2\pi i}\oint_C f(z)\,dz = \sum_{i=1}^{k} \operatorname*{Res}_{z=z_i} f(z). \tag{2.8}$$

We use the higher-order residue formula in [18, Chapter VI, Section 1].

For $\Re z > 1$ and $a > 0$, the Riemann and Hurwitz zeta functions are defined by

$$\zeta(z) = \sum_{n=1}^{\infty} n^{-z}, \qquad \zeta(z,a) = \sum_{n=0}^{\infty}(n+a)^{-z}.$$

Both functions admit meromorphic continuations, with a simple pole of residue one at $z = 1$. We write $\zeta'(z,a)$ for the derivative with respect to z. The special values used below are

$$\zeta(0,a) = \frac{1}{2} - a, \tag{2.9}$$

$$\zeta'(0,a) = \log\Gamma(a) - \frac{1}{2}\log(2\pi), \tag{2.10}$$

$$\zeta'(0) = -\frac{1}{2}\log(2\pi). \tag{2.11}$$

These identities follow from [4, p. 268] and [27, Eqs. (25.11.13), (25.11.18)].

For integers $1 \le \lambda \le \kappa$ and $\Re(s) < 0$, [4, Theorem 12.8] gives the functional equation

$$\begin{aligned}\zeta\Big(s, \frac{\lambda}{\kappa}\Big) =& 2\Gamma(1-s)(2\pi\kappa)^{s-1}\Big(\sin\frac{\pi s}{2}\sum_{1\le\nu\le\kappa}\cos\frac{2\pi\lambda\nu}{\kappa}\zeta\big(1-s,\frac{\nu}{\kappa}\big)\\ &+\cos\frac{\pi s}{2}\sum_{1\le\nu\le\kappa}\sin\frac{2\pi\lambda\nu}{\kappa}\zeta\big(1-s,\frac{\nu}{\kappa}\big)\Big).\end{aligned} \tag{2.12}$$

Hurwitz zeta functions satisfy the following two summation formulas, which can be seen in [7]. For a positive integer k and any $\theta = 1, 2, \ldots, k$,

$$\sum_{1\le\alpha\le k}\cos\frac{2\pi\alpha\theta}{k}\zeta\left(0,\frac{\alpha}{k}\right) = -\frac{1}{2}, \tag{2.13}$$

$$\sum_{1\le\alpha\le k}\cos\frac{2\pi\alpha\theta}{k}\zeta\left(2,\frac{\alpha}{k}\right) = \frac{\pi^2}{6}\left(6\theta^2 - 6k\theta + k^2\right). \tag{2.14}$$

The explicit Hurwitz-zeta estimates needed in Section 5 follow from the Euler–Maclaurin formula stated there; only the resulting compact-interval upper sums are evaluated by interval arithmetic.

The Mellin transformation [13] on a function $f(x)$ and its inversion form are defined as follows, with c in the strip of convergence and under the hypotheses of the Mellin inversion theorem:

$$\begin{aligned}\varphi(s) &= \{\mathcal{M}f\}(s) = \int_0^\infty x^{s-1}f(x)dx,\\ f(x) &= \{\mathcal{M}^{-1}\varphi\}(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}x^{-s}\varphi(s)ds =: \frac{1}{2\pi i}\int_{(c)}x^{-s}\varphi(s)ds.\end{aligned} \tag{2.15}$$

For an integer ν, the modified Bessel function of the first kind may be represented by

$$I_\nu(z) = \frac{1}{2\pi i}\int_{\mathcal{C}} w^{-\nu-1}\exp\left(\frac{z}{2}\left(w+\frac{1}{w}\right)\right)dw, \tag{2.16}$$

where $\mathcal{C}$ is a positively oriented simple loop around the origin. The Hankel deformation and the effective large-z remainder used later are recalled in Section 6; see [27, Sections 10.27 and 10.40] and [26, pp. 266–269].

3 Proof of the coprime case: Theorem 3.1

In this section, we are mainly concerned with the proof of Theorem 3.1. Firstly, we consider the calculation of $p_\ell(n)$ for $\ell = 4$ and a_i being replaced by $s, r-s, r+s, 2r-s$ for $1 \le i \le 4$, respectively, which gives

$$\sum_{n=0}^{\infty}p_4(n)q^n = \frac{1}{(1-q^s)(1-q^{r-s})(1-q^{r+s})(1-q^{2r-s})}. \tag{3.1}$$

Here the four parts $s, r-s, r+s, 2r-s$ are assumed to be pairwise coprime, and we put

$$\Delta = s(r-s)(r+s)(2r-s).$$

Theorem 3.1. *Let $1 \le s < r/2$, $(r,s) = 1$, and $k \ge 1$, and suppose that $s, r-s, r+s, 2r-s$ are pairwise coprime. The coefficient of q^n in*

$$\frac{\sum_{j=0}^{\infty}(-1)^j q^{r(j^2+(2k+1)j)/2-sj}(1-q^{2js+(2k+1)s})}{(1-q^s)(1-q^{r-s})(1-q^{r+s})(1-q^{2r-s})} \tag{3.2}$$

is positive for $n \ge L_1(r,s,k)$, where the exact, computable integer $L_1(r,s,k)$ is defined in (3.18).

Based on the residue theorem (2.8) and Lemma 2.1, we obtain the following expression for $p_4(n)$.

Proposition 3.2. *For $p_4(n)$ defined as above, we have that*

$$p_4(n) = P_3(n) + C_4(n), \tag{3.3}$$

where

$$P_3(n) = \frac{n^3 + 6n^2 r + n\left(21r^2 + 2rs - 2s^2\right)/2}{6\Delta}, \tag{3.4}$$

$C_4(n)$ is periodic with period Δ.

Proof. When $\ell = 4$, from (2.2) and (2.3), it follows that

$$\prod_{i=1}^{4} \frac{1}{1-q^{a_i}} = \sum_{i=1}^{4} \frac{c_i}{(1-q)^{5-i}} + \sum_{i=0}^{a_1a_2a_3a_4-1} \frac{b_i q^i}{1-q^{a_1a_2a_3a_4}}, \tag{3.5}$$

where c_i and b_i are constants. By using the residue theorem (2.8), we have that for $i = 1, 2, 3$,

$$c_i = \frac{(-1)^{i-1}}{(i-1)!} \lim_{q\to 1} \frac{\mathrm{d}^{i-1}}{\mathrm{d}q^{i-1}} \Big(\frac{(1-q)^4}{\prod_{i=1}^{4}(1-q^{a_i})} \Big).$$

Thus we obtain that

$$\begin{aligned}
c_1 &= \frac{1}{a_1a_2a_3a_4} \\
c_2 &= \frac{a_1+a_2+a_3+a_4-4}{2a_1a_2a_3a_4} \\
c_3 &= \frac{1}{12a_1a_2a_3a_4} \Big(\sum_{i=1}^{4} a_i(a_i-9) + \frac{3}{2}\sum_{i\ne j} a_i a_j + 14 \Big).
\end{aligned}$$

Taking the coefficient of q^n in the first sum on the right hand side of (3.5) and replacing a_i by $s, r-s, r+s, 2r-s$ for $1 \le i \le 4$, respectively, we see that

$$\begin{aligned}
P_3(n) + \kappa_4 &= \sum_{i=1}^{3} c_i \binom{n+4-i}{4-i} + c_4 \\
&= \frac{(n+1)\left(2n^2 + 12nr - 2n + 21r^2 + 2rs - 12r - 2s^2 + 2\right)}{12s(r-s)(2r-s)(r+s)} + c_4,
\end{aligned}$$

which leads to (3.4) by simplifying, with $\kappa_4 = r(5r^2 + 2rs - 2s^2)/6\Delta$. Moreover, by extracting the coefficients of q^n in the following sum

$$\sum_{i=0}^{a_1a_2a_3a_4-1} \frac{b_i q^i}{1-q^{a_1a_2a_3a_4}} = \sum_{i=0}^{a_1a_2a_3a_4-1} \sum_{k=0}^{\infty} b_i q^{i+(a_1a_2a_3a_4)k},$$

its coefficients are Δ-periodic. Thus, if $Q(n)$ denotes this periodic coefficient, then $C_4(n) := \kappa_4 + Q(n)$ is periodic, which completes the proof. □

We further consider the extrema of (3.3) by Lemma 2.2. Define

$$C_4^d := \min_{0\le h<\Delta} C_4(h), \qquad C_4^u := \max_{0\le h<\Delta} C_4(h) \tag{3.6}$$

and

$$\mathcal{B}_4 := \frac{1}{64} \sum_{a\in\{s,r-s,r+s,2r-s\}} a^2(a-1). \tag{3.7}$$

Lemma 3.3. *For any $n \ge 0$, we have*

$$P_3(n) + C_4^d \le p_4(n) \le P_3(n) + C_4^u. \tag{3.8}$$

Moreover, both extrema are rational and satisfy the entirely explicit bounds

$$\kappa_4 - \mathcal{B}_4 \le C_4^d \le C_4^u \le \kappa_4 + \mathcal{B}_4. \tag{3.9}$$

Proof. Put $\boldsymbol{a} = (s, r-s, r+s, 2r-s)$. In the notation of Lemma 2.2, the full contribution of $q = 1$ is $P_3(n) + \kappa_4$. Consequently,

$$C_4(n) = \kappa_4 + \mathscr{E}_{\boldsymbol{a}}(n).$$

Equations (2.5) and (3.7) now give $|C_4(n) - \kappa_4| \le \mathcal{B}_4$, which proves (3.9). Since C_4 is periodic, its minimum and maximum are attained on $0 \le h < \Delta$, and (3.8) follows. Finally, $C_4(h) = p_4(h) - P_3(h)$ is rational for every integer h, so both extrema are rational. □

Now, we are ready to prove Theorem 3.1.

Proof of Theorem 3.1. First, note that the numerator of (1.4) has the following parity decomposition

$$\begin{aligned}
&(-1)^k \sum_{j=k}^{\infty} (-1)^j q^{rj(j+1)/2} \left(q^{-sj} - q^{(j+1)s}\right) \\
&= q^{rk(k+1)/2-sk} \sum_{j=0}^{\infty} (-1)^j q^{r(j^2+(2k+1)j)/2-sj} (1 - q^{2js+(2k+1)s}) \\
&= q^{rk(k+1)/2-sk} \sum_{j=0}^{\infty} \left(q^{t_{1,j}} - q^{t_{2,j}} - q^{t_{3,j}} + q^{t_{4,j}}\right),
\end{aligned} \tag{3.10}$$

for which the above expression is obtained by considering the parities of j and

$$\begin{aligned}
t_{1,j} &= 2j^2r + 2jkr + jr - 2js, \\
t_{2,j} &= 2j^2r + 2jkr + jr + 2js + 2ks + s, \\
t_{3,j} &= 2j^2r + 2jkr + 3jr - 2js + kr + r - s, \\
t_{4,j} &= 2j^2r + 2jkr + 3jr + 2js + kr + 2ks + r + 2s.
\end{aligned}$$

Then assume that (3.2) can be expanded as follows

$$\sum_{n=0}^{\infty} C(n) q^n = \frac{\sum_{j=0}^{\infty} \left(q^{t_{1,j}} - q^{t_{2,j}} - q^{t_{3,j}} + q^{t_{4,j}}\right)}{(1-q^s)(1-q^{r-s})(1-q^{r+s})(1-q^{2r-s})}. \tag{3.11}$$

By applying the generating function (3.1) for $p_4(n)$, we can write the coefficients of q^n in (3.11) as follows

$$C(n) = \sum_{j=0}^{\infty} \Big(p_4(n - t_{1,j}) - p_4(n - t_{2,j}) - p_4(n - t_{3,j}) + p_4(n - t_{4,j})\Big). \tag{3.12}$$

In fact, there are only finitely many nonzero terms in (3.12) since $p_4(n) = 0$ if $n < 0$. We take the partial sum of $C(n)$ as follows

$$C(n,p) = \sum_{j=0}^{p-1} \Big(p_4(n - t_{1,j}) - p_4(n - t_{2,j}) - p_4(n - t_{3,j}) + p_4(n - t_{4,j}) \Big).$$

For $p \geq 1$ and $n \geq t_{4,p-1}$, applying (3.8) to the p complete blocks gives

$$\begin{aligned} C(n,p) \geq & \sum_{j=0}^{p-1} \Big(P_3(n - t_{1,j}) - P_3(n - t_{2,j}) - P_3(n - t_{3,j}) + P_3(n - t_{4,j}) \Big) \\ & - 2p(C_4^u - C_4^d) =: C_0(n,p). \end{aligned} \tag{3.13}$$

By noting that $1 \leq s < \frac{r}{2}$, it follows that $0 \leq t_{1,j} < t_{2,j} < t_{3,j} < t_{4,j} < t_{1,j+1}$. So all the nonnegative integers fall into the union of the following intervals

$$\bigcup_{p \geq 0} [t_{1,p}, t_{1,p+1}) = \bigcup_{p \geq 0} \Big[2p^2 r + 2pkr + pr - 2ps, 2(p+1)^2 r + 2(p+1)kr + (p+1)r - 2(p+1)s \Big).$$

Thus to prove the claim, it is sufficient to show that it holds for $n \in [2p^2 r + 2pkr + pr - 2ps, 2(p+1)^2 r + 2(p+1)kr + (p+1)r - 2(p+1)s)$. Then the proof of the claim can be decomposed into the following four cases.

Case 0: If $t_{4,p-1} \leq n < t_{1,p}$, then $C(n) = C(n,p) \geq C_0(n,p)$.

From Proposition 3.2 and (3.13), put $C_{0,d}(n,p) := C_0(n,p)$. Direct differentiation gives

$$\partial_n^2 C_{0,d}(n,p) = -\frac{2ps}{\Delta} < 0.$$

Thus it is enough to test $C_{0,d}(t_{4,p-1}, p) > 0$ and $\partial_n C_{0,d}(t_{1,p}, p) > 0$.

Case 1: If $t_{1,p} \leq n < t_{2,p}$, then we have

$$C(n) = C(n,p) + p_4(n - t_{1,p}).$$

Therefore

$$C(n) \geq C_{1,d}(n,p) := C_0(n,p) + P_3(n - t_{1,p}) + C_4^d. \tag{3.14}$$

Here $\partial_n C_{1,d}$ is a convex quadratic, since $\partial_n^3 C_{1,d} = 1/\Delta$. If $v_1(p)$ is its vertex, it is enough to show $C_{1,d}(t_{1,p}, p) > 0$ and $\partial_n C_{1,d}(v_1(p), p) > 0$.

Case 2: If $t_{2,p} \leq n < t_{3,p}$, we obtain that

$$C(n) = C(n,p) + p_4(n - t_{1,p}) - p_4(n - t_{2,p}),$$

and then,

$$C(n) \geq C_{2,d}(n,p) := C_0(n,p) + P_3(n - t_{1,p}) - P_3(n - t_{2,p}) + C_4^d - C_4^u. \tag{3.15}$$

Since

$$\partial_n^2 C_{2,d}(n,p) = \frac{2ps + 2ks + s}{\Delta} > 0,$$

it is enough to test $C_{2,d}(t_{2,p}, p) > 0$ and $\partial_n C_{2,d}(t_{2,p}, p) > 0$.

Case 3: If $t_{3,p} \leq n < t_{4,p}$, it follows that

$$C(n) = C(n,p) + p_4(n - t_{1,p}) - p_4(n - t_{2,p}) - p_4(n - t_{3,p}).$$

Thus,

$$C(n) \geq C_{3,d}(n,p) := C_0(n,p) + P_3(n-t_{1,p}) - P_3(n-t_{2,p}) - P_3(n-t_{3,p}) + C_4^d - 2C_4^u. \tag{3.16}$$

The quadratic $\partial_n C_{3,d}$ is concave, since $\partial_n^3 C_{3,d} = -1/\Delta$. Hence it is enough to verify

$$C_{3,d}(t_{3,p},p) > 0, \qquad \partial_n C_{3,d}(t_{3,p},p) > 0, \qquad \partial_n C_{3,d}(t_{4,p},p) > 0.$$

Put

$$\begin{aligned}\mathcal{C}_4 := \{&C_{0,d}(t_{4,p-1},p),\ C_{1,d}(t_{1,p},p),\ C_{2,d}(t_{2,p},p),\ C_{3,d}(t_{3,p},p),\\ &\partial_n C_{0,d}(t_{1,p},p),\ \partial_n C_{1,d}(v_1(p),p),\ \partial_n C_{2,d}(t_{2,p},p),\ \partial_n C_{3,d}(t_{3,p},p),\ \partial_n C_{3,d}(t_{4,p},p)\}.\end{aligned}$$

The first four members are rational polynomials in p of degree 4, each with positive leading coefficient $2kr^2s/\Delta$. The last five members are quadratic derivative tests with positive leading coefficients

$$\frac{2s}{\Delta}\big(r(k-1)+2s,\ r(k-1)+s,\ r(k-1)+2s,\ r(k+1)-2s, r(k+1)-2s\big).$$

For $f \in \mathcal{C}_4$, let $\rho(f)$ be its largest nonnegative real zero, with $\rho(f) = 0$ if no such zero exists, and set

$$p_{0,4} = 1 + \left\lfloor \max_{f\in\mathcal{C}_4} \rho(f) \right\rfloor. \tag{3.17}$$

The four cases therefore give

$$C(n) > 0 \qquad \big(n \geq t_{4,p_{0,4}-1} = (p_{0,4}+k)(2p_{0,4}r - r + 2s) =: L_1(r,s,k)\big). \tag{3.18}$$

By the definition of $p_{0,4}$ and the positivity of the displayed leading coefficients, every member of $\mathcal{C}_4$ is positive for $p \geq p_{0,4}$. This proves the theorem. □

4 Proof of the non-pairwise-coprime cases

In this section, we shall consider the restricted partitions with the four parts $s, r-s, r+s, 2r-s$ not pairwise coprime. More precisely, since $(r,s)=1$, the three parts $s, r-s, r+s$ are pairwise coprime when $2 \mid s$, while $s, r-s, 2r-s$ are pairwise coprime when $2 \nmid s$.

Theorem 4.1. *Let $1 \leq s < r/2$, $(r,s)=1$, $2 \mid s$, and $k \geq 1$. The coefficient of q^n in*

$$\frac{\sum_{j=0}^{\infty}(-1)^j q^{r(j^2+(2k+1)j)/2-sj}(1-q^{2js+(2k+1)s})}{(1-q^s)(1-q^{r-s})(1-q^{r+s})}$$

is positive for $n \geq L_2(r,s,k)$, where the exact, computable integer $L_2(r,s,k)$ is defined in (4.7).

Theorem 4.2. *Let $1 \leq s < r/2$, $(r,s)=1$, $2 \nmid s$, and $k \geq 1$. The coefficient of q^n in*

$$\frac{\sum_{j=0}^{\infty}(-1)^j q^{r(j^2+(2k+1)j)/2-sj}(1-q^{2js+(2k+1)s})}{(1-q^s)(1-q^{r-s})(1-q^{2r-s})}$$

is positive for $n \geq L_3(r,s,k)$, where the exact, computable integer $L_3(r,s,k)$ is defined in (4.13).

To prove Theorem 4.1 and Theorem 4.2, we shall alternatively consider the partition function with the following generating function

$$\sum_{n=0}^{\infty} p_{3,s}(n)q^n = \frac{1}{(1-q^s)(1-q^{r-s})(1-q^{r+s})}. \tag{4.1}$$

Denote $\Delta_1 = s(r-s)(r+s)$. Following the ideas given in the proof of Lemma 3.3, we derive the following result.

Lemma 4.3. *For $n \geq 0$, we have*

$$p_{3,s}(n) = P_{2,s}(n) + C_{3,s}(n), \qquad P_{2,s}(n) = \frac{n^2 + (2r+s)n + 2r + s - 1}{2\Delta_1}, \tag{4.2}$$

where $C_{3,s}(n)$ is Δ_1-periodic. Define

$$C_{3,s}^d := \min_{0 \leq h < \Delta_1} C_{3,s}(h), \qquad C_{3,s}^u := \max_{0 \leq h < \Delta_1} C_{3,s}(h).$$

Put

$$\mathcal{B}_{3,s} := \frac{1}{16} \sum_{a \in \{s, r-s, r+s\}} a(a-1).$$

Then

$$P_{2,s}(n) + C_{3,s}^d \leq p_{3,s}(n) \leq P_{2,s}(n) + C_{3,s}^u.$$

In addition, with

$$\kappa_e = \frac{5r^2 + 6rs - 12r - 6s + 6}{12\Delta_1},$$

$$\kappa_e - \mathcal{B}_{3,s} \leq C_{3,s}^d \leq C_{3,s}^u \leq \kappa_e + \mathcal{B}_{3,s}. \tag{4.3}$$

Proof. Apply Lemma 2.1 to the pairwise-coprime parts $s, r-s, r+s$. For three pairwise-coprime parts a, b, c, as in the proof of Proposition 3.2, the residue contribution from $q = 1$ is

$$\frac{n^2 + (a+b+c)n}{2abc} + \frac{3(a+b+c)^2 - (a^2+b^2+c^2)}{24abc}.$$

For the present three parts $s, r-s, r+s$, this is exactly $P_{2,s}(n) + \kappa_e$. Absorbing κ_e into the periodic term gives the stated decomposition (4.2). Lemma 2.2, applied to $\boldsymbol{a} = (s, r-s, r+s)$, gives $|C_{3,s}(n) - \kappa_e| \leq \mathcal{B}_{3,s}$ and hence (4.3) holds. □

Next, following the same procedure as in the proof of Theorem 3.1, we give the main steps to prove Theorem 4.1.

Proof of Theorem 4.1. In this proof for simplicity, we use $p_3(n)$ instead of $p_{3,s}(n)$ and similarly write P_2, C_3^u, C_3^d for $P_{2,s}, C_{3,s}^u, C_{3,s}^d$ when there is no confusion (namely when s is fixed). Consider the coefficient of q^n in the following series

$$\sum_{n=0}^{\infty} D(n) q^n = \frac{\sum_{j=0}^{\infty} (q^{t_{1,j}} - q^{t_{2,j}} - q^{t_{3,j}} + q^{t_{4,j}})}{(1-q^s)(1-q^{r-s})(1-q^{r+s})}. \tag{4.4}$$

Similarly, we first have

$$D(n) = \sum_{j=0}^{\infty} \big(p_3(n - t_{1,j}) - p_3(n - t_{2,j}) - p_3(n - t_{3,j}) + p_3(n - t_{4,j})\big). \tag{4.5}$$

Denote the first p complete blocks by $D(n,p)$ and put

$$D_0(n,p) := \sum_{j=0}^{p-1} \big(P_2(n - t_{1,j}) - P_2(n - t_{2,j}) - P_2(n - t_{3,j}) + P_2(n - t_{4,j})\big) - 2p(C_3^u - C_3^d). \tag{4.6}$$

Then $D(n,p) \geq D_0(n,p)$ whenever these p blocks are complete.

Then, let us consider the following four cases, using the exact lower functions defined below.

Case 0: If $t_{4,p-1} \leq n < t_{1,p}$, from (4.5), we obtain

$$D(n) = D(n,p) \geq D_{0,d}(n,p) := D_0(n,p).$$

Since $\partial_n D_{0,d} = -2ps/\Delta_1 < 0$, it is enough to show $D_{0,d}(t_{1,p}, p) > 0$.
Case 1: If $t_{1,p} \le n < t_{2,p}$, we see that

$$D(n) = D(n,p) + p_3(n - t_{1,p}).$$

Then

$$D(n) \ge D_{1,d}(n,p) := D_0(n,p) + P_2(n - t_{1,p}) + C_3^d.$$

This is a convex quadratic in n. Its vertex is

$$v_{D,1}(p) = -r + pr + 2kpr + 2p^2 r - s/2,$$

so it is enough to show $D_{1,d}(v_{D,1}(p), p) > 0$.
Case 2: If $t_{2,p} \le n < t_{3,p}$, it follows that

$$D(n) = D(n,p) + p_3(n - t_{1,p}) - p_3(n - t_{2,p}),$$

and then,

$$D(n) \ge D_{2,d}(n,p) := D_0(n,p) + P_2(n - t_{1,p}) - P_2(n - t_{2,p}) + C_3^d - C_3^u.$$

Since $\partial_n D_{2,d} = (2ps + 2ks + s)/\Delta_1 > 0$, it is enough to show $D_{2,d}(t_{2,p}, p) > 0$.
Case 3: If $t_{3,p} \le n < t_{4,p}$, we obtain

$$D(n) = D(n,p) + p_3(n - t_{1,p}) - p_3(n - t_{2,p}) - p_3(n - t_{3,p}).$$

Thereby,

$$D(n) \ge D_{3,d}(n,p) := D_0(n,p) + P_2(n - t_{1,p}) - P_2(n - t_{2,p}) - P_2(n - t_{3,p}) + C_3^d - 2C_3^u.$$

This is a concave quadratic in n, so it is enough to show

$$D_{3,d}(t_{3,p}, p) > 0, \qquad D_{3,d}(t_{4,p}, p) > 0.$$

Put

$$\mathcal{C}_s = \{D_{0,d}(t_{1,p}, p), \quad D_{1,d}(v_{D,1}(p), p), \quad D_{2,d}(t_{2,p}, p), \quad D_{3,d}(t_{3,p}, p), \quad D_{3,d}(t_{4,p}, p).\}$$

Their leading coefficients, in the same order, are

$$\frac{2s}{\Delta_1}\big(r(k-1) + 2s,\ r(k-1) + s,\ r(k-1) + 2s,\ r(k+1) - 2s,\ r(k+1) - 2s\big) > 0.$$

Let $p_{0,s}$ be one plus the floor of the largest nonnegative real zero among the polynomials in $\mathcal{C}_s$, taking that largest zero to be 0 if the set is empty. By the definition of the largest zeros and the positive leading coefficients, all five tests are positive for every integer $p \ge p_{0,s}$. Finally, we deduce that the coefficient of q^n is positive for

$$n \ge t_{4,p_{0,s}-1} = (p_{0,s} + k)(2p_{0,s} r - r + 2s) =: L_2(r, s, k). \tag{4.7}$$

This proves the theorem. □

We are going to consider the following partition function when $2 \nmid s$,

$$\sum_{n=0}^{\infty} E(n) q^n = \frac{\sum_{j=0}^{\infty} (q^{t_{1,j}} - q^{t_{2,j}} - q^{t_{3,j}} + q^{t_{4,j}})}{(1-q^s)(1-q^{r-s})(1-q^{2r-s})}. \tag{4.8}$$

Denote $\Delta_2 = s(r-s)(2r-s)$. Let $p_{3,r-s}(n)$ be defined by

$$\sum_{n=0}^{\infty} p_{3,r-s}(n) q^n = \frac{1}{(1-q^s)(1-q^{r-s})(1-q^{2r-s})}.$$

By replacing s by $r - s$ in Lemma 4.3, we can formally obtain the following comparison results of (4.2).

Lemma 4.4. *For $n \geq 0$, we have*

$$p_{3,r-s}(n) = P_{2,r-s}(n) + C_{3,r-s}(n), \qquad P_{2,r-s}(n) = \frac{n^2 + (3r-s)n + 3r - s - 1}{2\Delta_2}, \tag{4.9}$$

where $C_{3,r-s}(n)$ is Δ_2-periodic. Define

$$C^d_{3,r-s} := \min_{0 \leq h < \Delta_2} C_{3,r-s}(h), \qquad C^u_{3,r-s} := \max_{0 \leq h < \Delta_2} C_{3,r-s}(h).$$

Put

$$\mathcal{B}_{3,r-s} := \frac{1}{16} \sum_{a \in \{s, r-s, 2r-s\}} a(a-1).$$

Then

$$P_{2,r-s}(n) + C^d_{3,r-s} \leq p_{3,r-s}(n) \leq P_{2,r-s}(n) + C^u_{3,r-s}.$$

Furthermore, with

$$\kappa_o = \frac{11r^2 - 6rs - 18r + 6s + 6}{12\Delta_2},$$

$$\kappa_o - \mathcal{B}_{3,r-s} \leq C^d_{3,r-s} \leq C^u_{3,r-s} \leq \kappa_o + \mathcal{B}_{3,r-s}. \tag{4.10}$$

Proof. Apply Lemma 2.1 to the pairwise-coprime parts $s, r-s, 2r-s$. Their sum is $3r-s$ and their product is Δ_2, so the contribution from the pole at $q=1$ displayed in the proof of Lemma 4.3 is $P_{2,r-s}(n) + \kappa_o$. After this constant is absorbed, the remaining term is Δ_2-periodic. Applying Lemma 2.2 to $\boldsymbol{a} = (s, r-s, 2r-s)$ gives $|C_{3,r-s}(n) - \kappa_o| \leq \mathcal{B}_{3,r-s}$, which proves (4.10) and the asserted bounds. □

Proof of Theorem 4.2. For simplicity, we omit the subscript $r-s$ of $p_{3,r-s}(n)$, $P_{2,r-s}(n)$, $C^u_{3,r-s}$, $C^d_{3,r-s}$ when there is no confusion in this proof. Similar to the proof of Theorem 4.1, we first have

$$E(n) = \sum_{j=0}^{\infty} \Big(p_3(n - t_{1,j}) - p_3(n - t_{2,j}) - p_3(n - t_{3,j}) + p_3(n - t_{4,j}) \Big). \tag{4.11}$$

Denote the first p complete blocks by $E(n,p)$ and put

$$E_0(n,p) := \sum_{j=0}^{p-1} \Big(P_2(n - t_{1,j}) - P_2(n - t_{2,j}) - P_2(n - t_{3,j}) + P_2(n - t_{4,j}) \Big) - 2p(C^u_3 - C^d_3). \tag{4.12}$$

Then $E(n,p) \geq E_0(n,p)$.
Case 0: If $t_{4,p-1} \leq n < t_{1,p}$,

$$E(n) = E(n,p) \geq E_{0,d}(n,p) := E_0(n,p).$$

Since $\partial_n E_{0,d} = -2ps/\Delta_2 < 0$, it is enough to show $E_{0,d}(t_{1,p}, p) > 0$.
Case 1: If $t_{1,p} \leq n < t_{2,p}$,

$$E(n) = E(n,p) + p_3(n - t_{1,p}) \geq E_{1,d}(n,p),$$

where

$$E_{1,d}(n,p) := E_0(n,p) + P_2(n - t_{1,p}) + C^d_3.$$

This is a convex quadratic in n. If $v_{E,1}(p)$ denotes its vertex, it is enough to show $E_{1,d}(v_{E,1}(p), p) > 0$.
Case 2: If $t_{2,p} \leq n < t_{3,p}$,

$$E(n) = E(n,p) + p_3(n - t_{1,p}) - p_3(n - t_{2,p}) \geq E_{2,d}(n,p),$$

where

$$E_{2,d}(n,p) := E_0(n,p) + P_2(n - t_{1,p}) - P_2(n - t_{2,p}) + C^d_3 - C^u_3.$$

Since $\partial_n E_{2,d} = (2ps + 2ks + s)/\Delta_2 > 0$, it is enough to show $E_{2,d}(t_{2,p}, p) > 0$.
Case 3: If $t_{3,p} \le n < t_{4,p}$,

$$E(n) = E(n,p) + p_3(n - t_{1,p}) - p_3(n - t_{2,p}) - p_3(n - t_{3,p}) \ge E_{3,d}(n,p),$$

where

$$E_{3,d}(n,p) := E_0(n,p) + P_2(n - t_{1,p}) - P_2(n - t_{2,p}) - P_2(n - t_{3,p}) + C_3^d - 2C_3^u.$$

This is a concave quadratic in n, so it is enough to show $E_{3,d}(t_{3,p}, p) > 0$ and $E_{3,d}(t_{4,p}, p) > 0$.

Let $\mathcal{C}_{r-s}$ consist of $E_{3,d}(t_{3,p}, p)$, $E_{3,d}(t_{4,p}, p)$, $E_{0,d}(t_{1,p}, p)$, $E_{1,d}(v_{E,1}(p), p)$, and $E_{2,d}(t_{2,p}, p)$. These five quadratic polynomials in p have the same positive leading coefficients as in the even-s case, with Δ_1 replaced by Δ_2. Let $p_{0,r-s}$ be one plus the floor of their largest nonnegative real zero, again using 0 if there is no such zero. The definition of this largest zero and the positive leading coefficients then give

$$E(n) > 0 \qquad \left(n \ge t_{4,p_{0,r-s}-1} = (p_{0,r-s} + k)(2p_{0,r-s}r - r + 2s) =: L_3(r,s,k)\right). \tag{4.13}$$

This proves the theorem.

Following the same procedure as above, when $r = 2$ and $s = 1$ in (4.4), we obtain the following equivalent result with $R = 2S$.

Corollary 4.5. *For the positive integers $R = 2S$ and $k \ge 1$, the theta series*

$$(-1)^k \frac{\sum_{j=k}^{\infty} (-1)^j q^{Rj(j+1)/2} \left(q^{-Sj} - q^{(j+1)S}\right)}{(q^S, q^{R-S}; q^R)_\infty}$$

has nonnegative coefficients.

Proof. After replacing q^S by q, it is enough to take $r = 2$ and $s = 1$. Write

$$\frac{1}{(1-q)^2(1-q^3)} = \sum_{n \ge 0} u(n) q^n, \qquad u(n) = (L+1)(n+1) - \frac{3L(L+1)}{2}, \qquad L = \lfloor n/3 \rfloor,$$

and put $u(n) = 0$ for $n < 0$. Equivalently,

$$u(n) = \frac{n^2 + 5n}{6} + \varepsilon_\rho \quad (n \equiv \rho \pmod 3), \qquad (\varepsilon_0, \varepsilon_1, \varepsilon_2) = (1, 1, 2/3).$$

At $r = 2, s = 1$ the shifts satisfy

$$t_{1,j} = 4j(j+k), \qquad t_{2,j} = t_{3,j} = 4j^2 + 4kj + 4j + 2k + 1, \qquad t_{4,j} = t_{1,j+1}.$$

Thus the coefficient of the finite quotient is exactly

$$\sum_{j \ge 0} \left(u(n - t_{1,j}) - 2u(n - t_{2,j}) + u(n - t_{1,j+1})\right).$$

For $0 \le n < 2k + 1$, only the initial positive term occurs. If $n = 3m + \rho$, then

$$u(n) = \frac{(m+1)(3m + 2\rho + 2)}{2} > 0.$$

Put

$$P(n) = \frac{n^2 + 5n}{6}, \qquad x_j = 4j(j+k), \qquad y_j = 4j^2 + 4kj + 4j + 2k + 1.$$

For $p \ge 2$, let

$$S_p(n) = \sum_{j=0}^{p-1} \{P(n - x_j) - 2P(n - y_j) + P(n - x_{j+1})\}.$$

Since $2/3 \le \varepsilon_\rho \le 1$, the coefficient is bounded below on $x_p \le n < y_p$ and $y_p \le n < x_{p+1}$, respectively, by

$$L_1(n,p) = S_p(n) - \frac{2p}{3} + P(n - x_p) + \frac{2}{3}$$

and

$$L_3(n,p) = S_p(n) - \frac{2p}{3} + P(n - x_p) - 2P(n - y_p) - \frac{4}{3}.$$

The first is convex, with vertex $x_p + 2p - 5/2$ lying in $[x_p, y_p]$, and the second is concave. Direct substitution gives

$$\begin{aligned}
L_1(x_p, p) &= \frac{4kp^2}{3} + \left(\frac{4k^2}{3} - \frac{8}{3}\right)p + \frac{2}{3},\\
L_1(x_p + 2p - 5/2, p) &= \left(\frac{4k}{3} - \frac{2}{3}\right)p^2 + \left(\frac{4k^2}{3} - 1\right)p - \frac{3}{8},\\
L_3(y_p, p) &= \frac{4kp^2}{3} + \left(\frac{4k^2}{3} + \frac{4k}{3} + \frac{4}{3}\right)p + \frac{2k^2}{3} + \frac{7k}{3} - \frac{1}{3},\\
L_3(x_{p+1}, p) &= \frac{4kp^2}{3} + \left(\frac{4k^2}{3} + \frac{8k}{3} - \frac{8}{3}\right)p + \frac{4k^2}{3} + \frac{4k}{3} - \frac{10}{3}.
\end{aligned}$$

All four expressions are positive for $k \ge 1$ and $p \ge 2$; writing $k = K + 1$ and $p = P_0 + 2$ gives polynomials with nonnegative coefficients and positive constant terms. Hence all coefficients with $n \ge x_2 = 8k + 16$ are positive.

It remains to treat $2k + 1 \le n < 8k + 16$. Set

$$A = 2k + 1, \qquad B = 4k + 4, \qquad C = 6k + 9, \qquad D = 8k + 16.$$

On $[A, B)$, $[B, C)$, and $[C, D)$ the corresponding lower polynomials are

$$\begin{aligned}
F_1(n) &= P(n) - 2P(n - A) - \frac{4}{3},\\
F_2(n) &= P(n) - 2P(n - A) + 2P(n - B),\\
F_3(n) &= P(n) - 2P(n - A) + 2P(n - B) - 2P(n - C) - 2.
\end{aligned}$$

Their second derivatives are $-1/3, 1/3, -1/3$, respectively. The required endpoint and vertex values are

$$\begin{aligned}
F_1(A) &= \frac{2k^2}{3} + \frac{7k}{3} - \frac{1}{3}, & F_1(B) &= \frac{4k^2}{3} + \frac{4k}{3} - \frac{10}{3},\\
\min_{n \in \mathbb{R}} F_2(n) &= \frac{4k^2}{3} + \frac{4k}{3} - \frac{49}{24}, & F_3(C) &= 2k^2 + 5k + 1, & F_3(D) &= \frac{8k^2}{3} + \frac{16k}{3} - 6.
\end{aligned}$$

They are positive for $k \ge 2$ (write $k = K + 2$). For $k = 1$, the coefficients in degrees $0 \le n \le 23$ are, directly,

$$1, 2, 3, 3, 3, 3, 2, 1, 2, 2, 2, 4, 5, 6, 9, 9, 9, 11, 10, 9, 10, 8, 6, 6,$$

and are again positive. Thus the finite quotient has positive coefficients. Finally,

$$\frac{1}{(q;q^2)_\infty^2} = \frac{1}{(1-q)^2(1-q^3)} \frac{1}{(q^3, q^5; q^2)_\infty},$$

and the second factor has nonnegative coefficients. Multiplying by q^{k^2} and replacing q by q^S completes the proof. □

5 The nonmodular infinite products

In this section, based on Chern's method, we shall discuss the asymptotic expansion of the logarithm of the nonmodular product $G_{a,M}(q)$, namely $\Psi_{a,M}(q)+\Psi_{M-a,M}(q)$ as defined in (1.8). From this formula, we obtain the explicit bound for $G_{a,M}(q)$ on the minor arcs and an approximation near its dominant singularity $q=1$.

Put

$$\eta_M := \min\left\{\frac{\pi^2}{12M}, \frac{\pi^2(M-1)}{M^3}\right\}. \tag{5.1}$$

By using the circle method, we further extend Chern's idea to derive the following expansion for the logarithm $\Psi_{a,M}(q)+\Psi_{M-a,M}(q)$ of the nonmodular infinite product.

Theorem 5.1. *Let $X \geq 16$ be a positive number, and*

$$q := e^{-\tau+\frac{2\pi i h}{k_1}},$$

where $1 \leq h \leq k_1 \leq \lfloor\sqrt{2\pi X}\rfloor =: N$ with $(h,k_1)=1$ and $\tau := X^{-1}+2\pi i Y$ with $|Y| \leq 1/(k_1 N)$. Let a, M be positive integers with $M \geq 4$, $1 \leq a < M/2$, and $(a,M)=1$. Denote by b the unique integer between 1 and (k_1,M) such that $b \equiv -ha \pmod{(k_1,M)}$. Then

$$\Psi_{a,M}(q)+\Psi_{M-a,M}(q) = \frac{\pi^2}{\tau}\frac{(k_1,M)^2}{k_1^2 M}\left(\frac{2b(b-(k_1,M))}{(k_1,M)^2}+\frac{1}{3}\right)+2E_1,$$

where $2E_1$ is the error term. More precisely,

$$2|\Re E_1| \leq 30M^{3/2}X^{1/2}+2X^{1/2}\log X+2M^2. \tag{5.2}$$

5.1 Proof of Theorem 5.1

In this subsection, we prove the expression for $\Psi_{a,M}(q)+\Psi_{M-a,M}(q)$ as given in Theorem 5.1.

Proof of Theorem 5.1. We first use Chern's method to derive the main and error terms of $\Psi_{a,M}(q)+\Psi_{M-a,M}(q)$. Recall that for the given k_1, we take

$$q := e^{-\tau+\frac{2\pi i h}{k_1}},$$

where $1 \leq h \leq k_1 \leq \lfloor\sqrt{2\pi X}\rfloor =: N$ with $(h,k_1)=1$ and $\tau := X^{-1}+2\pi i Y$ with $|Y| \leq 1/(k_1 N)$. Denote

$$K := k_1\frac{M}{(k_1,M)} = k_1\frac{M}{M^*},$$

where $M^* = (k_1,M)$. From the definition (1.8), one can see that

$$\begin{aligned}
\Psi_{a,M}(q) &= \log\left(\frac{1}{(q^{a+2M};q^M)_\infty}\right)\\
&= \log\left(\frac{1}{(q^a;q^M)_\infty}\right)-\log\left(\frac{1}{(1-q^a)(1-q^{a+M})}\right)\\
&= \sum_{\substack{m\geq 1\\ m\equiv a \bmod M}}\sum_{\ell\geq 1}\frac{q^{\ell m}}{\ell}-\left(\sum_{\ell\geq 1}\frac{q^{a\ell}}{\ell}+\sum_{\ell\geq 1}\frac{q^{(a+M)\ell}}{\ell}\right)\\
&=: \Phi_{a,M}(q)-T_a\\
&= \sum_{\substack{1\leq\lambda\leq K\\ \lambda\equiv a \bmod M}}\sum_{1\leq\mu\leq k_1} e^{2\pi i h\mu\lambda/k_1}\sum_{r,t\geq 0}\frac{e^{-(rk_1+\mu)(tK+\lambda)\tau}}{rk_1+\mu}-T_a,
\end{aligned}$$

where the last equality follows by substituting ℓ and m as follows:

$$\ell = rk_1 + \mu \quad \text{for} \quad 1 \le \mu \le k_1,$$
$$m = tK + \lambda \quad \text{for} \quad 1 \le \lambda \le K,\ \lambda \equiv a \pmod{M}.$$

Here $T_a = \sum_{\ell \ge 1}(q^{a\ell} + q^{(a+M)\ell})/\ell$. Applying the Mellin transform (2.15) to $\varphi(s) = \Gamma(s)$ and $f(x) = e^{-x}$ and substituting x by $(rk_1 + \mu)(tK + \lambda)\tau$, we have

$$\sum_{r,t \ge 0} \frac{e^{-(rk_1+\mu)(tK+\lambda)\tau}}{rk_1 + \mu} = \frac{1}{2\pi i}\int_{(3/2)} \frac{\Gamma(s)}{\tau^s k_1^{s+1} K^s} \zeta\left(s, \frac{\lambda}{K}\right)\zeta\left(1+s, \frac{\mu}{k_1}\right) ds.$$

By employing (2.12), (2.6), and $e^{i\theta} = \cos\theta + i\sin\theta$, we obtain that

$$\begin{aligned}
&\Phi_{a,M}(q) = \Psi_{a,M}(q) + T_a \\
&= \frac{1}{4\pi i k_1 K} \sum_{\substack{1\le\lambda\le K \\ \lambda\equiv a \bmod M}} \sum_{\substack{1\le\mu\le k_1 \\ 1\le\nu\le K}} \cos\frac{2\pi\mu h\lambda}{k_1}\cos\frac{2\pi\nu\lambda}{K}\int_{(3/2)} \frac{\zeta(1+s,\mu/k_1)\zeta(1-s,\nu/K)}{z^s\cos(\pi s/2)} ds \\
&\quad + \frac{1}{4\pi i k_1 K} \sum_{\substack{1\le\lambda\le K \\ \lambda\equiv a \bmod M}} \sum_{\substack{1\le\mu\le k_1 \\ 1\le\nu\le K}} \cos\frac{2\pi\mu h\lambda}{k_1}\sin\frac{2\pi\nu\lambda}{K}\int_{(3/2)} \frac{\zeta(1+s,\mu/k_1)\zeta(1-s,\nu/K)}{z^s\sin(\pi s/2)} ds \\
&\quad + \frac{1}{4\pi k_1 K} \sum_{\substack{1\le\lambda\le K \\ \lambda\equiv a \bmod M}} \sum_{\substack{1\le\mu\le k_1 \\ 1\le\nu\le K}} \sin\frac{2\pi\mu h\lambda}{k_1}\sin\frac{2\pi\nu\lambda}{K}\int_{(3/2)} \frac{\zeta(1+s,\mu/k_1)\zeta(1-s,\nu/K)}{z^s\sin(\pi s/2)} ds \\
&\quad + \frac{1}{4\pi k_1 K} \sum_{\substack{1\le\lambda\le K \\ \lambda\equiv a \bmod M}} \sum_{\substack{1\le\mu\le k_1 \\ 1\le\nu\le K}} \sin\frac{2\pi\mu h\lambda}{k_1}\cos\frac{2\pi\nu\lambda}{K}\int_{(3/2)} \frac{\zeta(1+s,\mu/k_1)\zeta(1-s,\nu/K)}{z^s\cos(\pi s/2)} ds \\
&=: \gamma_{1,-s} - \gamma_{2,-s} - \gamma_{3,-s} + \gamma_{4,-s},
\end{aligned}$$

where $z := \tau k_1/(2\pi)$. As given in [10, Eq. (2.19)], replacing s by $-s$, letting ρ be the unique integer in $[1, k_1]$ such that $\rho \equiv -h\lambda \pmod{k_1}$, reversing the direction of the integration path, and shifting the path back to (3/2) by using the residue theorem (2.8), it can be seen that

$$\Phi_{a,M}(q) = \Psi_{a,M}(q) + T_a = \gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 - 2\pi i(R_1 + R_2 + R_3 + R_4),$$

where γ_j denotes the transformed contour integral arising from the term $\gamma_{j,-s}$ fixed above and R_j is the sum of the residues of the corresponding integrand inside the strip $-3/2 < \Re(s) < 3/2$. Thus

$$\Psi_{a,M}(q) = \gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 - 2\pi i(R_1 + R_2 + R_3 + R_4) - T_a.$$

We will give a brief proof which will show that the main terms for $\Psi_{a,M}(q)$ come from R_1, R_4, while the others are the error terms.

First, notice that

$$\left|\Re\sum_{\ell\ge1}\frac{q^{c\ell}}{\ell}\right| \le \sum_{\ell\ge1}\frac{e^{-c\ell/X}}{\ell} = -\log(1-e^{-c/X}) \le \log\frac{X}{c} + \frac{c}{X} \qquad (c>0).$$

Since $1 - e^{-u} \ge ue^{-u}$ for $u > 0$, summing over the four deleted exponents gives

$$|\Re(T_a + T_{M-a})| \le 4\log X - \log\Delta_{a,M} + \frac{4M}{X}. \tag{5.3}$$

Let $\gamma_j^{(c)}$ denote the jth complete transformed integral above with the residue class a replaced by $c \in \{a, M-a\}$. The estimates obtained in [10, Theorem 3.2], applied separately for $c = a$ and $c = M-a$, give

$$\left|\Re\sum_{j=1}^{4}\gamma_j^{(c)}\right| \le 13.02\frac{M^{3/2}}{(M^*)^{5/2}}X^{1/2} + \frac{3e^{0.28\pi^2M^*/M}}{(e^{0.28\pi^2M^*/M}-1)^2}.$$

Summing these two inequalities, using $M^* \ge 1$, and noting that $e^u/(e^u-1)^2$ decreases for $u > 0$, gives

$$\left|\Re\sum_{c\in\{a,M-a\}}\sum_{j=1}^{4}\gamma_j^{(c)}\right| \le 26.04M^{3/2}X^{1/2} + \frac{6e^{0.28\pi^2/M}}{(e^{0.28\pi^2/M}-1)^2}. \tag{5.4}$$

Next, we shall give the accurate value or the estimation of each R_i for $1 \le i \le 4$. By the definition of γ_1, the residues of the integrand in γ_1 are given as follows:

$$\begin{aligned}\mathcal{R}_1 &:= \sum_{|\Re(s)|<3/2}\operatorname*{Res}_s\frac{\zeta(1-s,\mu/k_1)\zeta(1+s,\nu/K)}{z^{-s}\cos(\pi s/2)}\\ &= \frac{2\zeta(2,\mu/k_1)\zeta(0,\nu/K)}{\pi z} - \frac{2z\zeta(0,\mu/k_1)\zeta(2,\nu/K)}{\pi} - \log z\\ &\quad - \frac{\Gamma'}{\Gamma}\left(\frac{\mu}{k_1}\right) + \frac{\Gamma'}{\Gamma}\left(\frac{\nu}{K}\right) =: \mathcal{R}_{11}+\mathcal{R}_{12}+\mathcal{R}_{13}+\mathcal{R}_{14}.\end{aligned} \tag{5.5}$$

Thus

$$R_1 = \frac{1}{4\pi i k_1 K}\sum_{\substack{1\le\lambda\le K\\ \lambda\equiv a \bmod M}}\sum_{\substack{1\le\mu\le k_1\\ 1\le\nu\le K}}\cos\frac{2\pi\mu\rho}{k_1}\cos\frac{2\pi\nu\lambda}{K}\mathcal{R}_1.$$

Here for brevity, we only give two parts R_{11} and R_{12} of R_1 in detail as an illustration. Furthermore, with (2.13) and (2.14), it can be deduced that

$$\begin{aligned}R_{11} &= \frac{1}{z}\frac{1}{2i\pi^2k_1K}\sum_{\substack{1\le\lambda\le K\\ \lambda\equiv a \bmod M}}\sum_{1\le\mu\le k_1}\cos\frac{2\pi\mu\rho}{k_1}\zeta\left(2,\frac{\mu}{k_1}\right)\sum_{1\le\nu\le K}\cos\frac{2\pi\nu\lambda}{K}\zeta\left(0,\frac{\nu}{K}\right)\\ &= \frac{1}{z}\frac{1}{2i\pi^2k_1K}\sum_{\substack{1\le\rho\le k_1\\ \rho\equiv b \bmod M^*}}\frac{\pi^2}{6}(6\rho^2-6k_1\rho+k_1^2)\left(-\frac{1}{2}\right)\\ &= \frac{1}{z}\frac{1}{2i\pi^2k_1K}\sum_{j=0}^{k_1/M^*-1}\frac{\pi^2}{6}\big(6(b+jM^*)^2-6k_1(b+jM^*)+k_1^2\big)\left(-\frac{1}{2}\right).\end{aligned}$$

For the sum,

$$\sum_{0\le j<k_1/M^*}\big(6(b+jM^*)^2-6k_1(b+jM^*)+k_1^2\big) = \frac{k_1}{M^*}\big(6b^2-6bM^*+(M^*)^2\big). \tag{5.6}$$

Together with $z = \tau k_1/(2\pi)$ this gives $-2\pi i R_{11}^{(a)}$ in (5.8). For $\mathcal{R}_{12}$, by (2.13) and (2.14), it can be seen that

$$\begin{aligned}R_{12} &= -\frac{z}{2i\pi^2k_1K}\sum_{\substack{1\le\lambda\le K\\ \lambda\equiv a \bmod M}}\sum_{1\le\mu\le k_1}\cos\frac{2\pi\mu\rho}{k_1}\zeta\left(0,\frac{\mu}{k_1}\right)\sum_{1\le\nu\le K}\cos\frac{2\pi\nu\lambda}{K}\zeta\left(2,\frac{\nu}{K}\right)\\ &= \frac{\tau}{48i\pi M}(6a^2-6aM+M^2).\end{aligned}$$

Thus

$$|\Re(-2\pi i R_{12})| \le \frac{M}{24X}. \tag{5.7}$$

Put

$$b^* := \begin{cases} M^* - b, & b < M^*, \\ M^*, & b = M^*. \end{cases}$$

Following the same procedure as deducing R_{11} and R_{12}, one can directly deduce R_{ij} as follows; see [10, Section 4] for details. For the residue class a, the two terms needed for the principal part are

$$-2\pi i R_{11}^{(a)} = \frac{1}{\tau}\frac{\pi^2}{6k_1^2 M}(6b^2 - 6bM^* + (M^*)^2), \tag{5.8}$$

$$-2\pi i R_{41}^{(a)} = \begin{cases} \dfrac{2\pi i (M^*)^2}{\tau k_1^2 M}\left\{-\zeta'\left(-1, \dfrac{b}{M^*}\right) + \zeta'\left(-1, \dfrac{b^*}{M^*}\right)\right\}, & b < M^*, \\ 0, & b = M^*. \end{cases} \tag{5.9}$$

The second equation is Chern's evaluation [10, Eq. (4.37)].

The other exceptional terms $R_{42}^{(a)}$ and $R_{42}^{(M-a)}$ cancel after the two residue classes are paired. Choose h' with $hh' \equiv -1 \pmod{k_1}$ and put $\mathcal{R}_a = \{1 \le \rho < k_1 : \rho \equiv b \pmod{M^*}\}$. For $\rho \in \mathcal{R}_a$, the congruences

$$\ell_\rho^{(a)} \equiv a \pmod{M}, \qquad \ell_\rho^{(a)} \equiv h'\rho \pmod{k_1}$$

determine a unique $\ell_\rho^{(a)} \in [1, K]$. Chern's exact formula [10, Eq. (4.41)] is

$$R_{42}^{(a)} = \frac{\tau}{48\pi K}\sum_{\rho \in \mathcal{R}_a}\left(6(\ell_\rho^{(a)})^2 - 6K\ell_\rho^{(a)} + K^2\right)\cot\frac{\pi\rho}{k_1}. \tag{5.10}$$

For the residue class $M - a$, the involution $(\rho, \ell_\rho^{(a)}) \mapsto (k_1 - \rho, K - \ell_\rho^{(a)})$ preserves the quadratic factor and changes the sign of the cotangent. Consequently

$$R_{42}^{(a)} + R_{42}^{(M-a)} = 0. \tag{5.11}$$

The main terms come from R_{11} and R_{41}. To avoid confusion, let $-2\pi i R_{11}^{(a)}$ and $-2\pi i R_{41}^{(a)}$ denote the main terms of $\Psi_{a,M}$ and the corresponding $b = b(a)$ satisfying $b(a) \equiv -ha \pmod{M^*}$. It can be seen that

$$b(M-a) \equiv -h(M-a) \equiv ha \pmod{M^*},$$

so $b(M-a) = b^*(a)$ and $b(a) = b^*(M-a)$. It follows that

$$-2\pi i\big(R_{11}^{(a)} + R_{41}^{(a)} + R_{11}^{(M-a)} + R_{41}^{(M-a)}\big) = \frac{1}{\tau}\frac{(M^*)^2\pi^2}{k_1^2 M}\left\{\frac{2b(b-M^*)}{(M^*)^2} + \frac{1}{3}\right\}. \tag{5.12}$$

Indeed, the two R_{41} terms cancel after b and b^* are interchanged, whereas the two quadratic R_{11} terms add to the expression displayed. The deleted initial factors in (5.3) and the remaining residues are included in $2E_1$. For either residue class, the following table bounds the absolute real part of $-2\pi i$ times the term in its left column. These bounds follow from Chern's (4.11), (4.12), (4.13), (4.24)–(4.26), and (4.35) [10]; $R_{13} = 0$ because the residue class is strictly less than M:

R_{12}	$M/(24X)$
R_{13}	0
R_{14}	$M/2$
R_{21}	$0.44X^{1/2}\log X + 1.3X^{1/2} + 0.25\log X + 0.75$
R_{22}	$0.5\log X + 0.92$
R_{23}	$0.25\log X + M/2 + (3/2)\log M + 2.59$
R_3	0.

Since $\Delta_{a,M} \geq 1$ and $X \geq 16$, we have $2M + 49M/(12X) \leq 433M/192 < 7M/3$. With (5.4), we then have

$$2|\Re E_1| \leq 26.04M^{3/2}X^{1/2} + \frac{7}{3}M + \frac{6e^{0.28\pi^2/M}}{(e^{0.28\pi^2/M}-1)^2} + 3\log M + 2.6X^{1/2} + (0.88X^{1/2}+6)\log X + 9. \tag{5.13}$$

If $u = 0.28\pi^2/M$, then

$$\frac{e^u}{(e^u-1)^2} = \frac{1}{4\sinh^2(u/2)} \leq \frac{1}{u^2}.$$

In particular, $6/u^2 = 3750M^2/(49\pi^4) < M^2$. For $X \geq 16$ and $M \geq 4$, we also have $5/2 \leq \log X \leq X^{1/2}$, $\log M \leq M/2$, and $M^{3/2} \geq 8$. Therefore

$$2.6X^{1/2} + 6\log X + 9 \leq 8.77X^{1/2} + 9 < 3.96M^{3/2}X^{1/2}$$

Also,

$$\frac{7}{3}M + 3\log M \leq \frac{23}{6}M < M^2, \qquad 0.88X^{1/2}\log X < 2X^{1/2}\log X.$$

Together with $6e^u/(e^u-1)^2 < M^2$, these inequalities reduce (5.13) directly to (5.2). This completes the proof of Theorem 5.1.

□

5.2 The bound for $G_{a,M}(q)$ on the minor arcs

We next aim to discuss the values of $G_{a,M}(q)$ on the minor arcs, i.e. when q is away from the $q = 1$ major arc. By using the above asymptotic results for $\Psi_{a,M}(q) + \Psi_{M-a,M}(q)$, we obtain the following bound for $G_{a,M}(q)$.

Theorem 5.2. *Let M, a be positive integers with $M \geq 4$, $1 \leq a < M/2$, and $(a, M) = 1$, and let $X \geq 110M^7$. If $t \in \mathbb{R}$ and*

$$q = e^{-1/X+2\pi it}, \qquad \frac{1}{2\pi X} \leq \|t\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{1}{2},$$

where $\|t\|_{\mathbb{R}/\mathbb{Z}} := \min_{m\in\mathbb{Z}} |t-m|$, then

$$|G_{a,M}(q)| \leq \exp\left\{\left(\frac{\pi^2}{3M} - \eta_M\right)X\right\}. \tag{5.14}$$

Proof. Let $\mathcal{M}_G(b)$ denote the main term of $\Psi_{a,M}(q) + \Psi_{M-a,M}(q)$. Then, by Theorem 5.1, we have

$$\mathcal{M}_G(b) = \frac{\pi^2}{\tau}\frac{(M^*)^2}{k_1^2 M}\left(\frac{2b(b-M^*)}{(M^*)^2} + \frac{1}{3}\right),$$

$$2E_1 = \log G_{a,M}(q) - \mathcal{M}_G(b).$$

Choose a Farey fraction h/k_1 of order $N = \lfloor\sqrt{2\pi X}\rfloor$ and the representative of t modulo 1 such that $t \equiv h/k_1 - Y \pmod 1$ and $|Y| \leq 1/(k_1 N)$. Put $M^* = (k_1, M)$, and let b be as in Theorem 5.1. The Farey arcs cover the full circle $|q| = e^{-1/X}$, so this choice is available for every t in the stated range. Moreover,

$$\frac{\Re(\tau^{-1})}{X} = \frac{1}{1+(2\pi XY)^2} \leq 1. \tag{5.15}$$

The coefficient of $\pi^2 X/M$ in $\Re\mathcal{M}_G(b)$ is

$$\frac{\Re(\tau^{-1})}{X}\frac{(M^*)^2}{k_1^2}\left\{\frac{2b(b-M^*)}{(M^*)^2} + \frac{1}{3}\right\}. \tag{5.16}$$

On the complementary part of the 1/1 arc, $|Y| = \|t\|_{\mathbb{R}/\mathbb{Z}} \geq 1/(2\pi X)$, so (5.15) is at most 1/2 and (5.16) is at most 1/6. If $k_1 \geq 2$ and $M^* = 1$, it is at most $1/(3k_1^2) \leq 1/12$. Finally, if $M^* > 1$, then $1 \leq b \leq M^* - 1$, since $(ha, M^*) = 1$, and convexity in b gives

$$\frac{(M^*)^2}{k_1^2}\left\{\frac{2b(b-M^*)}{(M^*)^2} + \frac{1}{3}\right\} \leq \frac{(M^*)^2/3 - 2M^* + 2}{k_1^2}.$$

For $2 \leq M^* \leq 4$, the numerator $(M^*)^2/3 - 2M^* + 2$ is negative, so (5.16) is nonpositive. For $M^* \geq 5$, this numerator is positive. Using (5.15), $k_1 \geq M^*$, $M^* \leq M$, and the monotonicity of $1/3 - 2/x + 2/x^2$ for $x \geq 2$, we obtain the upper bound $1/3 - 2/M + 2/M^2$ for (5.16). Together with (5.1), these three cases give

$$\Re \mathcal{M}_G(b) \leq \frac{\pi^2 X}{3M} - 2\eta_M X. \tag{5.17}$$

To estimate $|\Re E_1|$ in (5.2), we put $\mathcal{E}_M(X) = 30M^{3/2}X^{1/2} + 2X^{1/2}\log X + 2M^2$. Each term in

$$\frac{\mathcal{E}_M(X)}{X} = 30M^{3/2}X^{-1/2} + 2X^{-1/2}\log X + 2M^2X^{-1}$$

decreases for $X \geq 110M^7$. The function $M^{-3/2}\log(110M^7)$ decreases for $M \geq 4$, since its derivative is $M^{-5/2}(7 - \frac{3}{2}\log(110M^7)) < 0$. At $X = 110M^7$,

$$\begin{aligned}\frac{M^2\mathcal{E}_M(110M^7)}{110M^7} &= \frac{30}{\sqrt{110}} + \frac{2\log(110M^7)}{\sqrt{110}\,M^{3/2}} + \frac{1}{55M^3} \\ &\leq \frac{30}{\sqrt{110}} + \frac{\log(110\cdot 4^7)}{4\sqrt{110}} + \frac{1}{3520} < \frac{\pi^2}{3},\end{aligned}$$

where we used $\sqrt{110} > 52/5$ and $\log(110 \cdot 4^7) < 15$. Since

$$M^2\eta_M = \min\left\{\frac{\pi^2 M}{12}, \frac{\pi^2(M-1)}{M}\right\} \geq \frac{\pi^2}{3} \qquad (M \geq 4),$$

we obtain

$$\mathcal{E}_M(X) \leq \eta_M X \qquad (M \geq 4,\ X \geq 110M^7). \tag{5.18}$$

Combining (5.17) and (5.18) with the relation

$$\log|G_{a,M}(q)| = \Re \log G_{a,M}(q) \leq \Re\mathcal{M}_G(b) + 2|\Re E_1|,$$

we complete the proof. □

5.3 Approximations of $G_{a,M}(q)$ near the poles

In this subsection, we shall discuss the approximation of $G_{a,M}(q)$ near its dominant singularity $q = 1$. On the complementary minor arcs we use Theorem 5.2.

Theorem 5.3. *Let M, a be positive integers with $M \geq 4$, $1 \leq a < M/2$, and $(a, M) = 1$. If $\tau = X^{-1} + 2\pi iY$ with $X \geq 1$ and $|Y| \leq 1/(2\pi X)$, then we have*

$$\log G_{a,M}(e^{-\tau}) = \frac{\pi^2}{3M\tau} + 4\log\tau - \log\left(2\sin\frac{a\pi}{M}\right) + \log\Delta_{a,M} + E_+, \tag{5.19}$$

where $\Delta_{a,M} = a(M-a)(M+a)(2M-a)$, $\log G_{a,M}(e^{-\tau})$ is determined by its absolutely convergent logarithmic series and $\log\tau$ is the principal branch, and

$$|E_+| \leq 5.5M^{3/4}X^{-3/4}.$$

Proof. By using the definition of $G_{a,M}(q)$ in (1.7) and applying the Mellin transform (2.15), we obtain that

$$\begin{aligned}\log G_{a,M}(e^{-\tau}) &= \sum_{m\geq 2}\left(\log\frac{1}{1-e^{-(mM+a)\tau}}+\log\frac{1}{1-e^{-(mM+M-a)\tau}}\right)\\ &= \sum_{m\geq 2}\sum_{\ell\geq 1}\left(\frac{e^{-(mM+a)\ell\tau}}{\ell}+\frac{e^{-(mM+M-a)\ell\tau}}{\ell}\right)\\ &= \frac{1}{2\pi i}\int_{(3/2)}\frac{\Gamma(s)\zeta(s+1)}{(M\tau)^s}\left\{\zeta\left(s,\frac{a}{M}\right)+\zeta\left(s,\frac{M-a}{M}\right)\right\}ds\\ &\quad -\frac{1}{2\pi i}\int_{(3/2)}\frac{\Gamma(s)\zeta(s+1)}{\tau^s}\left\{a^{-s}+(M-a)^{-s}+(M+a)^{-s}+(2M-a)^{-s}\right\}ds.\end{aligned}$$

Let

$$\begin{aligned}G_1(s) &:= \frac{\Gamma(s)\zeta(s+1)}{(M\tau)^s}\left\{\zeta\left(s,\frac{a}{M}\right)+\zeta\left(s,\frac{M-a}{M}\right)\right\},\\ G_2(s) &:= \frac{\Gamma(s)\zeta(s+1)}{\tau^s}\left\{a^{-s}+(M-a)^{-s}+(M+a)^{-s}+(2M-a)^{-s}\right\}.\end{aligned}$$

Then, by changing the paths of integration to $\Re(s)=-3/4$ and considering the residues in the strip $-3/4<\Re(s)<3/2$, it follows that

$$\log G_{a,M}(e^{-\tau}) = \operatorname*{Res}_{s=1}G_1(s)+\operatorname*{Res}_{s=0}\{G_1(s)-G_2(s)\}+\frac{1}{2\pi i}\int_{(-3/4)}\Big(G_1(s)-G_2(s)\Big)ds.$$

The function G_1 has a simple pole at $s=1$ and at most a simple pole at $s=0$, since $\zeta(0,a/M)+\zeta(0,(M-a)/M)=0$. Using (2.9)–(2.11), we obtain

$$\operatorname*{Res}_{s=1}G_1 = \frac{1}{M\tau}\Gamma(1)\zeta(2)(1+1)=\frac{\pi^2}{3M\tau},$$

and

$$\begin{aligned}\operatorname*{Res}_{s=0}G_1 &= -\Big(\frac{1}{2}-\frac{a}{M}+\frac{1}{2}-\frac{M-a}{M}\Big)\log M\tau+\zeta'\Big(0,\frac{a}{M}\Big)+\zeta'\Big(0,\frac{M-a}{M}\Big)\\ &= -\log\Big(2\sin\frac{a\pi}{M}\Big).\end{aligned}$$

Similarly, G_2 has a double pole at $s=0$, and we obtain

$$\begin{aligned}\operatorname*{Res}_{s=0}G_2 &= -4\log\tau+\log\frac{1}{a}+\log\frac{1}{M-a}+\log\frac{1}{M+a}+\log\frac{1}{2M-a}\\ &= -4\log\tau-\log\big(a(M-a)(M+a)(2M-a)\big).\end{aligned}$$

The minus sign preceding G_2 therefore gives $+4\log\tau+\log\Delta_{a,M}$ in (5.19). Thus

$$\begin{aligned}E_+ &= \frac{1}{2\pi i}\int_{(-3/4)}\big(G_1(s)-G_2(s)\big)\,ds,\\ |E_+| &\leq \frac{1}{2\pi}\left|\int_{(-3/4)}G_1(s)\,ds\right|+\frac{1}{2\pi}\left|\int_{(-3/4)}G_2(s)\,ds\right|.\end{aligned}\tag{5.20}$$

Since $|Y|\leq 1/(2\pi X)$,

$$|\arg\tau|\leq\frac{\pi}{4},\qquad |\tau|\leq\sqrt{2}X^{-1},\qquad |\tau^{-s}|\leq|\tau|^{3/4}e^{\pi|\Im(s)|/4}\quad(\Re(s)=-3/4).$$

Put

$$\mathscr{K}(t) := e^{\pi|t|/4}\left|\Gamma\left(-\frac{3}{4}+it\right)\zeta\left(\frac{1}{4}+it\right)\right|.$$

For $0 \le t \le 5$, let

$$H_0(t) := \frac{2\Gamma(7/4)\zeta(7/4)}{(2\pi)^{7/4}}\cosh\frac{\pi t}{2};$$

the Hurwitz functional equation [27, Eq. (25.11.9)] gives $|\zeta(-3/4+it,x)| \le H_0(t)$ uniformly for $0 < x < 1$. For $t \ge 5$, put $N_t = 1 + \lfloor t \rfloor$ and define the majorant obtained from the Euler–Maclaurin formula [27, Eqs. (25.11.4) and (25.11.6)] by

$$\begin{aligned} H_\infty(t) := {} & \sum_{\nu=1}^{N_t}\nu^{3/4} + \frac{(N_t+1)^{7/4}}{|-7/4+it|} + \frac{1}{2}(N_t+1)^{3/4} \\ & + \left\{\frac{|-3/4+it|}{12} + \frac{|(-3/4+it)(1/4+it)|}{3}\right\} N_t^{-1/4}. \end{aligned} \tag{5.21}$$

The absolute remainder in the Euler–Maclaurin formula [27, Section 25.11(iii)] gives $|\zeta(-3/4+it,x)| \le H_\infty(t)$ uniformly for $0 < x < 1$. The zeta functional equation [27, Eq. (25.4.2)] and Euler's reflection formula also give

$$\mathscr{K}(t) = \frac{(2\pi)^{1/4}e^{\pi t/4}}{\sqrt{t^2+9/16}}\left|\frac{\sin(\pi/8+i\pi t/2)}{\sin(\pi/4+i\pi t)}\right|\left|\zeta\left(\frac{3}{4}-it\right)\right| \qquad (t \ge 0). \tag{5.22}$$

For $w = 3/4 - it$, the Euler–Maclaurin formula with $N_0 = 32$ reads

$$\begin{aligned} \zeta(w) &= \sum_{m=1}^{31} m^{-w} + \frac{32^{1-w}}{w-1} + \frac{1}{2}\,32^{-w} + \frac{w}{12}\,32^{-w-1} + R_{32}(w), \\ |R_{32}(w)| &\le \frac{|w(w+1)|}{21\,32^{7/4}}. \end{aligned} \tag{5.23}$$

The routine `MellinMajorantCheck[]` applies interval arithmetic to (5.21)–(5.23) on $[0,40]$ and certifies

$$\frac{2}{\pi}\int_0^5 \mathscr{K}H_0 < \frac{23}{20}, \qquad \frac{2}{\pi}\int_5^{40}\mathscr{K}H_\infty < \frac{7}{50}, \qquad \frac{1}{\pi}\int_0^{40}\mathscr{K} < \frac{37}{50}. \tag{5.24}$$

For the tails, if $t \ge 40$, the same Euler–Maclaurin remainder, now with $N_1 = \lceil t \rceil$, gives

$$\begin{aligned} \left|\zeta\left(\frac{3}{4}-it\right)\right| &< \left(5+\frac{1}{20}+\frac{1}{80}+\frac{1}{400}+\frac{1}{10}\right)t^{1/4} < 8t^{1/4}, \\ H_\infty(t) &< \left(1+\frac{1}{20}+\frac{3}{100}+\frac{1}{400}+\frac{3}{8}\right)t^{7/4} < 4t^{7/4}. \end{aligned}$$

These estimates follow term by term from $t \le N_1 \le t+1 \le 41t/40$ and $t \le N_t \le t+1$, $N_t + 1 \le 21t/20$ in (5.21) and (5.23). Moreover, the identity $|\sin(\alpha+i\beta)|^2 = \sin^2\alpha + \sinh^2\beta$ gives

$$\left|\frac{\sin(\pi/8+i\pi t/2)}{\sin(\pi/4+i\pi t)}\right| \le \frac{\cosh(\pi t/2)}{\sinh(\pi t)} = \frac{e^{-\pi t/2}}{1-e^{-\pi t}}.$$

Hence, with $c = 157/200 < \pi/4$, equation (5.22) implies $\mathscr{K}(t) < 16t^{-3/4}e^{-ct}$; here one may use $(2\pi)^{1/4} < 8/5$ and $(1-e^{-\pi t})^{-1} < 5/4$. Therefore

$$\frac{2}{\pi}\int_{40}^{\infty}\mathscr{K}(t)H_\infty(t)\,dt < \frac{128}{\pi}e^{-40c}\left(\frac{40}{c}+\frac{1}{c^2}\right) < \frac{1}{100},$$

$$\frac{1}{\pi}\int_{40}^{\infty} \mathscr{K}(t)\,dt < \frac{16}{\pi c}\,40^{-3/4}e^{-40c} < \frac{1}{100}. \tag{5.25}$$

Combining the compact interval certificates with (5.25) yields

$$\left|\frac{1}{2\pi i}\int_{(-3/4)} G_1(s)ds\right| < \frac{13}{10}|M\tau|^{3/4}, \qquad \left|\frac{1}{2\pi i}\int_{(-3/4)} G_2(s)ds\right| < \frac{147}{50}|M\tau|^{3/4}. \tag{5.26}$$

The first inequality of (5.26) follows from (5.24) and (5.25). For the second bound, writing $x = a/M \in (0, 1/2)$ gives

$$x^{3/4} + (1-x)^{3/4} + (1+x)^{3/4} + (2-x)^{3/4} < \frac{391}{100};$$

the left-hand side increases to its endpoint value at $x = 1/2$. Together with

$$\frac{1}{\pi}\int_0^{\infty} \mathscr{K}(t)dt < \frac{3}{4},$$

this gives $(391/100)(3/4) < 147/50$.

Summing the two bounds in (5.26) and using $|\tau| \le \sqrt{2}X^{-1}$, we have

$$2^{3/8}\left(\frac{13}{10} + \frac{147}{50}\right) < 5.5.$$

This completes the proof. □

6 The asymptotic formula and Theorem 1.4

The circle method was first used by Hardy and Ramanujan [17] in 1918 to investigate the asymptotic growth of the partition function $p(n)$, which was later refined to deal with the coefficients of the q-expansions of modular forms [29]. Recently, Chern [10] extended this method to study the nonmodular infinite products. The idea is to express the coefficient as an integral over a circle and then deform the contour to obtain an asymptotic formula.

For the analytic estimates below, let M, a be positive integers with $M \ge 4$, $1 \le a < M/2$, and $(a, M) = 1$, and let n be a positive integer. We put

$$X = \sqrt{\frac{3Mn}{\pi^2}}.$$

Put

$$A_M := \frac{\pi^2}{3M}, \qquad \xi := A_M X,$$

and define the two thresholds

$$\begin{aligned} X_{\mathrm{p}}(M) &:= \max\{110M^7, 1.2\times 10^7 M\}, & T_{\mathrm{p}}(M) &:= \left\lceil \frac{10X_{\mathrm{p}}(M)^2}{3M} \right\rceil, \\ X_{\mathrm{s}}(M) &:= \max\{110M^7, 2\times 10^{15} M\}, & T_{\mathrm{s}}(M) &:= \left\lceil \frac{10X_{\mathrm{s}}(M)^2}{3M} \right\rceil. \end{aligned} \tag{6.1}$$

Since $\pi^2 < 10$, the implication $n \ge T_\nu(M) \Rightarrow X \ge X_\nu(M)$ holds for $\nu \in \{\mathrm{p}, \mathrm{s}\}$.

6.1 Proof of Theorem 1.5

We use Theorems 5.2 and 5.3 to estimate $g_{a,M}(n)$ for sufficiently large n.

Proof of Theorem 1.5. Assume first that $X \geq 110M^7$. We integrate along the circle $|q| = e^{-1/X}$ in the positive direction, using the parametrization

$$q = e^{-1/X+2\pi it}, \qquad -\frac{1}{2\pi X} \leq t \leq 1 - \frac{1}{2\pi X}.$$

As t increases through this interval of length 1, the point q traverses the circle once counterclockwise, and $dq = 2\pi iq\,dt$. The residue theorem (2.8) therefore gives

$$\begin{aligned} g_{a,M}(n) &= \frac{1}{2\pi i}\int_{|q|=e^{-\frac{1}{X}}} \frac{G_{a,M}(q)}{q^{n+1}}dq \\ &= e^{\frac{n}{X}} \int_{-\frac{1}{2\pi X}}^{1-\frac{1}{2\pi X}} G_{a,M}\left(e^{-1/X+2\pi it}\right) e^{-2\pi int}\,dt. \end{aligned}$$

Let I denote the interval $\left[-\frac{1}{2\pi X}, 1-\frac{1}{2\pi X}\right]$ and denote

$$\begin{aligned} I_1 &:= \left[-\frac{1}{2\pi X}, \frac{1}{2\pi X}\right], \\ I_2 &:= \left[\frac{1}{2} - \frac{1}{2\pi X}, \frac{1}{2} + \frac{1}{2\pi X}\right], \\ I_3 &:= I \setminus (I_1 \cup I_2) = \left(\frac{1}{2\pi X}, \frac{1}{2} - \frac{1}{2\pi X}\right) \cup \left(\frac{1}{2} + \frac{1}{2\pi X}, 1 - \frac{1}{2\pi X}\right]. \end{aligned}$$

Then

$$g_{a,M}(n) = \sum_{i=1}^{3} e^{\frac{n}{X}} \int_{I_i} G_{a,M}\left(e^{-1/X+2\pi it}\right) e^{-2\pi int}\,dt =: \sum_{i=1}^{3} g_i(n). \tag{6.2}$$

For the convolutions in the next subsection, we also estimate

$$J_{c,M}(q) := \frac{1}{(q^{2M+c}, q^{2M-c}; q^M)_\infty} = \sum_{n\geq 0} J_{c,M}(n)q^n,$$

where $c \in \{a, M-a\}$. Put $F_{a,M}(q) := (q^a, q^{M-a}; q^M)_\infty^{-1}$. The three products can be written as

$$\begin{aligned} G_{a,M} &= F_{a,M}(1-q^a)(1-q^{M-a})(1-q^{M+a})(1-q^{2M-a}), \\ J_{a,M} &= F_{a,M}(1-q^a)(1-q^{M-a})(1-q^{M+a}), \\ J_{M-a,M} &= F_{a,M}(1-q^a)(1-q^{M-a})(1-q^{2M-a}). \end{aligned} \tag{6.3}$$

Then

$$J_{a,M} = \frac{G_{a,M}}{1-q^{2M-a}}, \qquad J_{M-a,M} = \frac{G_{a,M}}{1-q^{M+a}}. \tag{6.4}$$

For $U \in \{G_{a,M}, J_{a,M}, J_{M-a,M}\}$ let m be the number of finite factors and let $\mathcal{P}_U$ be the product of their exponents:

U	m	$\mathcal{P}_U$
$G_{a,M}$	4	$a(M-a)(M+a)(2M-a)$
$J_{a,M}$	3	$a(M-a)(M+a)$
$J_{M-a,M}$	3	$a(M-a)(2M-a)$.

Write $\Lambda_U := \mathcal{P}_U/(2\sin(a\pi/M))$ and $e_M(X) := 5.5M^{3/4}X^{-3/4}$. For $U = G_{a,M}$, Theorem 5.3 gives

$$U(e^{-\tau}) = \Lambda_U \tau^m \exp\{A_M/\tau + E_U(\tau)\}, \qquad |E_U(\tau)| \le e_M(X). \tag{6.5}$$

For $U = J_{a,M}$ or $J_{M-a,M}$, the Mellin representation in the proof of Theorem 5.3 has the same term $G_1(s)$, while $G_2(s)$ contains only the three deleted exponents in (6.3). Its residue at zero is $-3\log\tau - \log\mathcal{P}_U$. The termwise absolute bound for the remaining integral is no larger than the four-term bound in (5.26). Thus (6.5) holds for all three products with the same bound $e_M(X)$. For $t \in I_2 \cup I_3$, the point $q = e^{-1/X+2\pi it}$ lies on the minor arcs of Theorem 5.2. Since $X \ge 110M^7$, we have $1 < c \le 2M \le X$ for $c \in \{M+a, 2M-a\}$. The inequality $1 - e^{-y} \ge y/2$ for $0 \le y \le 1$ gives

$$|1 - q^c| \ge 1 - |q|^c = 1 - e^{-c/X} \ge \frac{c}{2X} > \frac{1}{2X}.$$

Using the two quotient identities (6.4), Theorem 5.2 therefore gives

$$|U(e^{-1/X+2\pi it})| \le 2X \exp\{(A_M - \eta_M)X\}. \tag{6.6}$$

Write $U(q) = \sum_{n\ge0} u(n)q^n$, and define $I_j^U(n)$ from the three intervals above by replacing $G_{a,M}$ with U in (6.2). On I_1, put $\tau = X^{-1} - 2\pi it$. As t increases from $-1/(2\pi X)$ to $1/(2\pi X)$, the point τ moves from $X^{-1} + iX^{-1}$ to $X^{-1} - iX^{-1}$. Since $dt = -d\tau/(2\pi i)$, reversing the endpoints gives

$$\begin{aligned} I_1^U(n) &= \frac{1}{2\pi i}\int_{X^{-1}-iX^{-1}}^{X^{-1}+iX^{-1}} U(e^{-\tau})e^{n\tau}\,d\tau \\ &= \frac{\Lambda_U}{2\pi i}\int_{X^{-1}-iX^{-1}}^{X^{-1}+iX^{-1}} \tau^m e^{A_M/\tau+n\tau}e^{E_U(\tau)}\,d\tau. \end{aligned}$$

Define

$$I_1^{U(0)}(n) := \frac{\Lambda_U}{2\pi i}\int_{X^{-1}-iX^{-1}}^{X^{-1}+iX^{-1}} \tau^m e^{A_M/\tau+n\tau}\,d\tau.$$

We complete this vertical segment by two horizontal rays to form the Hankel contour

$$C: \quad -\infty - iX^{-1} \longrightarrow X^{-1} - iX^{-1} \longrightarrow X^{-1} + iX^{-1} \longrightarrow -\infty + iX^{-1}.$$

We use the contour representation (2.16) and let $\tau = \sqrt{A_M/n}\,w = X^{-1}w$. Then C becomes

$$C^*: \quad -\infty - i \longrightarrow 1 - i \longrightarrow 1 + i \longrightarrow -\infty + i.$$

Consequently,

$$\begin{aligned} H_U(n) &:= \frac{\Lambda_U}{2\pi i}\int_C \tau^m e^{A_M/\tau+n\tau}\,d\tau \\ &= \Lambda_U\left(\frac{A_M}{n}\right)^{(m+1)/2}\frac{1}{2\pi i}\int_{C^*} w^m \exp\left(\sqrt{A_M n}\left(w + \frac{1}{w}\right)\right)dw \\ &= \Lambda_U\left(\frac{A_M}{n}\right)^{(m+1)/2} I_{-m-1}(2\sqrt{A_M n}), \end{aligned} \tag{6.7}$$

where I_ν is the modified Bessel function. The contour decomposition gives

$$I_1^{U(0)}(n) - H_U(n) = -\frac{\Lambda_U}{2\pi i}\left(\int_{-\infty-iX^{-1}}^{X^{-1}-iX^{-1}} + \int_{X^{-1}+iX^{-1}}^{-\infty+iX^{-1}}\right)\tau^m e^{A_M/\tau+n\tau}\,d\tau.$$

Thus

$$u(n) - H_U(n) = \{I_1^U - I_1^{U(0)}\} + \{I_1^{U(0)} - H_U\} + I_2^U + I_3^U. \tag{6.8}$$

We first estimate the error on the vertical segment. For $\tau = X^{-1} + iy$, $|y| \le X^{-1}$, one has

$$|\tau| \le \frac{\sqrt{2}}{X}, \qquad \Re\frac{A_M}{\tau} \le A_M X, \qquad \left|e^{E_U(\tau)} - 1\right| \le e^{e_M(X)} - 1.$$

With $n/X = A_M X$, it follows that

$$|I_1^U - I_1^{U(0)}| \le \frac{\Lambda_U 2^{m/2}}{\pi X^{m+1}} e^{2A_M X}(e^{e_M(X)} - 1). \tag{6.9}$$

For the lower and upper horizontal rays of C, we use $\tau = x - iX^{-1}$ and $\tau = x + iX^{-1}$, respectively. The first ray is traversed with x increasing from $-\infty$ to X^{-1}, and the second with x decreasing from X^{-1} to $-\infty$. On both rays,

$$\Re\frac{A_M}{x \pm iX^{-1}} = \frac{A_M x}{x^2 + X^{-2}} \le \frac{A_M X}{2}, \qquad |e^{n(x \pm iX^{-1})}| = e^{nx},$$

and

$$|x \pm iX^{-1}|^m \le \begin{cases} (-x + X^{-1})^m, & x \le 0, \\ 2^m X^{-m}, & 0 \le x \le X^{-1}. \end{cases}$$

The integral over the negative half-line satisfies

$$\begin{aligned} \int_{-\infty}^{0} e^{nx}(-x + X^{-1})^m \, dx &= \sum_{\ell=0}^{m} \binom{m}{\ell} X^{-(m-\ell)} \int_0^\infty e^{-nx} x^\ell \, dx \\ &= \sum_{\ell=0}^{m} \binom{m}{\ell} \frac{\ell!}{X^{m-\ell} n^{\ell+1}}, \end{aligned}$$

whereas

$$\int_0^{X^{-1}} 2^m X^{-m} e^{nx} \, dx = \frac{2^m(e^{n/X} - 1)}{X^m n} \le \frac{2^m e^{n/X}}{X^m n}.$$

Each ray is bounded by the sum of these two integrals multiplied by $\Lambda_U e^{A_M X/2}/(2\pi)$. Adding the bounds for the two rays gives

$$|I_1^{U(0)} - H_U| \le \frac{\Lambda_U e^{A_M X/2}}{\pi} \left\{ \sum_{\ell=0}^{m} \binom{m}{\ell} \frac{\ell!}{X^{m-\ell} n^{\ell+1}} + \frac{2^m e^{n/X}}{X^m n} \right\}. \tag{6.10}$$

For I_2, put $t = 1/2 + v$ and then $\tau = X^{-1} - 2\pi i v$. This gives the corresponding vertical integral near $q = -1$:

$$I_2^U(n) = \frac{(-1)^n}{2\pi i} \int_{X^{-1} - iX^{-1}}^{X^{-1} + iX^{-1}} U(-e^{-\tau}) e^{n\tau} \, d\tau.$$

The interval I_3 gives the two remaining arcs of the same circle. Both I_2 and I_3 are covered by the minor-arc estimate, and $|I_2 \cup I_3| = 1 - 1/(\pi X) < 1$. Applying (6.6) on these intervals gives

$$|I_2^U + I_3^U| \le e^{n/X} \int_{I_2 \cup I_3} |U(e^{-1/X + 2\pi i t})| \, dt < 2X \exp\left(2A_M X - \eta_M X\right). \tag{6.11}$$

Normalize these estimates by

$$\mathfrak{u}_U(n) := \frac{\mathcal{P}_U}{4\sqrt{\pi}\sin(a\pi/M)} A_M^{m/2+1/4} n^{-m/2-3/4} e^{2\sqrt{A_M n}}. \tag{6.12}$$

For $m \in \{3, 4\}$ put

$$\nu := m + 1, \qquad z := 2\sqrt{A_M n} = 2A_M X, \qquad A_1(\nu) := \frac{4\nu^2 - 1}{8}, \qquad B_\nu := \nu^2 - \frac{1}{4} = 2A_1(\nu).$$

Since $\nu \in \{4,5\}$, [27, Eq. (10.27.1)] gives $I_{-\nu}(z) = I_\nu(z)$, and direct simplification of (6.7) and (6.12) gives

$$\frac{H_U(n)}{\mathfrak{u}_U(n)} = e^{-z}\sqrt{2\pi z}\, I_\nu(z).$$

Let $R_+ := R_1(\nu, ze^{\pi i})$ and $R_0 := R_1(\nu, z)$ denote the remainders in [27, Eq. (10.40.10)] with $\ell = 1$. The upper-sign instance of [27, Eq. (10.34.3)], with the continuation index there equal to zero, yields

$$e^{-z}\sqrt{2\pi z}\, I_\nu(z) - 1 = R_+ - ie^{-\nu\pi i}e^{-2z}(1 + R_0).$$

By [27, Eqs. (10.40.11)–(10.40.12)] and $\chi(1) = \pi/2$,

$$|R_+| \le \frac{\pi A_1(\nu)}{z}\exp\!\left(\frac{\pi B_\nu}{2z}\right), \qquad |R_0| \le \frac{2A_1(\nu)}{z}\exp\!\left(\frac{B_\nu}{z}\right).$$

Here $(A_1(4), B_4) = (63/8, 63/4)$ and $(A_1(5), B_5) = (99/8, 99/4)$. For $z \ge 10^6$, the inequalities $\pi < 355/113$ and $e^y < (1-y)^{-1}$ for $0 < y < 1$ give

$$z|R_+| < \frac{(355/113)(99/8)}{1 - (355/113)(99/8)10^{-6}} < 39.$$

Also, with $y := 99/(4z)$,

$$1 + |R_0| \le 1 + ye^y < \frac{1}{1-y} < 2.$$

Since $e^{2z} > 2z$, it follows that $e^{-2z}(1 + |R_0|) < 1/z$. Consequently,

$$\frac{|H_U(n) - \mathfrak{u}_U(n)|}{\mathfrak{u}_U(n)} < \frac{40}{z} < \frac{100}{z} \qquad (z \ge 10^6). \tag{6.13}$$

When $X \ge 110M^7$ and $X/M \ge 1.2 \times 10^7$, we have $e_M(X) < 1/5$ and hence $e^{e_M(X)} - 1 \le (5/4)e_M(X)$. Since $n = A_M X^2$, the normalizing factor can also be written as

$$\mathfrak{u}_U(n) = \frac{\Lambda_U}{2\sqrt{\pi}} A_M^{-1/2} X^{-m-3/2} e^{2A_M X}.$$

Dividing (6.9) by $\mathfrak{u}_U(n)$ gives

$$\frac{|I_1^U - I_1^{U(0)}|}{\mathfrak{u}_U(n)} \le \frac{2^{m/2+1}}{\sqrt{\pi}}\sqrt{\xi}\,(e^{e_M(X)} - 1) \le 55\sqrt{\frac{\pi}{3}}\left(\frac{M}{X}\right)^{1/4} < 58\left(\frac{M}{X}\right)^{1/4}.$$

(6.10) gives

$$\frac{|I_1^{U(0)} - H_U|}{\mathfrak{u}_U(n)} \le \frac{2}{\sqrt{\pi\xi}} e^{-3\xi/2}\left\{\sum_{\ell=0}^{m}\binom{m}{\ell}\frac{\ell!}{\xi^\ell} + 2^m e^\xi\right\} \le \frac{1296}{\xi^2}.$$

Here $m \le 4$ and $\xi \ge 1$, so the sum is at most 65 and we also used $2^m \le 16$ and $e^{\xi/2} \ge \xi^2/8$. Finally, $\Lambda_U \ge 1/2$, $A_M < 1$, and $X \ge 1$ give

$$\frac{|I_2^U + I_3^U|}{\mathfrak{u}_U(n)} < \frac{4\sqrt{\pi}}{\Lambda_U}\sqrt{A_M}\, X^{m+5/2} e^{-\eta_M X} \le 80X^7 e^{-\eta_M X}.$$

Thus (6.8) yields the uniform bound

$$\frac{|u(n) - H_U(n)|}{\mathfrak{u}_U(n)} \le 58\left(\frac{M}{X}\right)^{1/4} + \frac{1296}{\xi^2} + 80X^7 e^{-\eta_M X}. \tag{6.14}$$

Each term on the right decreases throughout the ranges in (6.1). For the last term this follows from

$$\eta_M \geq \frac{\pi^2}{3M^2} > \frac{3}{M^2}, \qquad \frac{d}{dX}\log(X^7 e^{-\eta_M X}) = \frac{7}{X} - \eta_M < 0 \quad (X \geq 110M^7),$$

and the other two terms, as well as $100/z$, are decreasing. Put $\epsilon = 1/500000$. For $X \geq X_{\mathrm{p}}(M)$, since $z = 2\xi > 6X/M$, one has

$$\frac{100}{z} < \epsilon, \qquad \frac{1296}{\xi^2} < \epsilon.$$

Moreover, at $X = 110M^7$ the inequality $e^y \geq y^{11}/11!$ gives

$$80X^7 e^{-\eta_M X} < \frac{80 \cdot 110^7\, 11!}{330^{11} M^6} < \epsilon.$$

Finally, direct calculation gives

$$58\left(\frac{M}{X}\right)^{1/4} \leq \frac{58}{(1.2\times 10^7)^{1/4}} < \frac{493}{500} \qquad (X \geq X_{\mathrm{p}}(M)),$$
$$58\left(\frac{M}{X}\right)^{1/4} \leq \frac{58}{(2\times 10^{15})^{1/4}} < \frac{217}{25000} \quad (X \geq X_{\mathrm{s}}(M)).$$

Thus

$$\frac{493}{500} + 3\epsilon < \frac{99}{100}, \qquad \frac{217}{25000} + 3\epsilon < \frac{1}{100}.$$

The sum of $100/z$ and the right side of (6.14) is therefore less than 0.99 at $X = X_{\mathrm{p}}(M)$ and less than 0.01 at $X = X_{\mathrm{s}}(M)$, uniformly for $M \geq 4$. Hence

$$\begin{aligned} &0.01\,\mathfrak{u}_U(n) \leq u(n) \leq 1.99\,\mathfrak{u}_U(n) \quad (n \geq T_{\mathrm{p}}(M)),\\ &0.99\,\mathfrak{u}_U(n) \leq u(n) \leq 1.01\,\mathfrak{u}_U(n) \quad (n \geq T_{\mathrm{s}}(M)). \end{aligned} \tag{6.15}$$

The errors in (6.13) and (6.14) tend to zero for fixed M. For $U = G_{a,M}$, $m = 4$ and $\mathcal{P}_U = a(M-a)(M+a)(2M-a)$, so $g_{a,M}(n) \sim H_U(n)$ is precisely the asymptotic formula in Theorem 1.5. This completes the proof. □

Proof of Theorem 1.6. For $U = G_{a,M}$, the second line of (6.15), together with $m = 4$ and $\mathcal{P}_U = a(M-a)(M+a)(2M-a)$ in (6.12), gives the claimed two-sided bound for $n \geq T_{\mathrm{s}}(M)$. Finally, the proof is complete by replacing a, M by s, r, respectively. □

6.2 Merca's stronger conjecture for sufficiently large n

Note that from Corollary 4.5, Merca's stronger conjecture 1.2 has been certified for the cases $R = 2S$. Moreover, the authors [12] have proved the cases when $R = 3S$. It therefore remains to consider the reduced modulus $r \geq 4$; in the analytic estimates below, $M = r \geq 4$. Now, we are ready to prove the stronger version of Merca's conjecture on the truncated Jacobi triple theta series (1.5) for sufficiently large n. Here the analytic parameters are $M = r$ and $a = s$.

Proof of Theorem 1.4. With the factorization (1.6), together with the finite estimates in Sections 3 and 4, we can reduce all three cases to one convolution. Write

case	$b(n)$	$U(q)$	n_b
four pairwise-coprime parts	$C(n)$	$G_{s,r}(q)$	$\lceil L_1(r,s,k)\rceil$
non-pairwise-coprime, $2 \mid s$	$D(n)$	$J_{s,r}(q)$	$\lceil L_2(r,s,k)\rceil$
non-pairwise-coprime, $2 \nmid s$	$E(n)$	$J_{r-s,r}(q)$	$\lceil L_3(r,s,k)\rceil$,

(6.16)

where L_1, L_2, L_3 are defined in (3.18), (4.7), and (4.13). Put

$$\delta_0 := \frac{rk(k+1)}{2} - sk.$$

If $H(n)$ denotes the coefficient of the series in (1.5) with $R = r$ and $S = s$, then (1.6) gives the exact identity

$$\sum_{n\ge 0} H(n)q^n = q^{\delta_0}\Big(\sum_{j\ge 0} b(j)q^j\Big)U(q) \tag{6.17}$$

in each row of (6.16). The results in Sections 3 and 4 give $b(n) \ge 1$ for $n \ge n_b$. Define

$$N_b := \sum_{0\le j<n_b} \max\{0, -b(j)\}. \tag{6.18}$$

For the product U in the selected row, use the values of m and $\mathcal{P}_U$ above with $M = r$ and $a = s$, and set

$$B = \frac{2\pi}{\sqrt{3r}}, \qquad \gamma = \frac{m}{2} + \frac{3}{4}, \qquad T = T_{\mathrm{p}}(r), \qquad c_U = \frac{\mathcal{P}_U}{4\sqrt{\pi}\sin(s\pi/r)} A_r^{m/2+1/4}.$$

Then $\mathfrak{u}_U(n) = c_U n^{-\gamma} e^{B\sqrt{n}}$, $0 < B \le \pi/\sqrt{3}$, and $\gamma \in \{9/4, 11/4\}$. Also $u(n) \ge 0$ for every $n \ge 0$, since U is a reciprocal product of factors $1 - q^j$. For

$$V(n) := n^{-\gamma} e^{B\sqrt{n}}, \qquad K(B,\gamma) := \left\lceil 796\, 2^{\gamma} e^{B} \right\rceil,$$

suppose that

$$0.01 c_U V(n) \le u(n) \le 1.99 c_U V(n) \qquad (n \ge T),$$

where $c_U > 0$, $0 < \gamma \le 11/4$, and $0 < B \le \pi/\sqrt{3}$. Then we claim that

$$\sum_{j=0}^{n} b(j)u(n-j) > 0 \quad \text{for} \quad n \ge N_b^* := \max\{16, 4n_b^2, 2T, (K(B,\gamma)N_b + 1)^2\}. \tag{6.19}$$

For $0 \le j \le \sqrt{n}$, $n - j \ge n - \sqrt{n} \ge n/2 \ge T$. Moreover,

$$\frac{V(n-j)}{V(n)} = \left(\frac{n}{n-j}\right)^{\gamma} \exp\left(B(\sqrt{n-j} - \sqrt{n})\right) \le 2^{\gamma}.$$

Thus the absolute value of the total negative contribution to $\sum_{j=0}^{n} b(j)u(n-j)$ is at most

$$1.99 c_U 2^{\gamma} N_b V(n). \tag{6.20}$$

For the positive contribution, put $L = \left\lceil \frac{\sqrt{n}}{2} \right\rceil$ and $R = \lfloor \sqrt{n} \rfloor$. Since $n \ge 16$,

$$R - L + 1 \ge \frac{\sqrt{n}}{2} - 1 \ge \frac{\sqrt{n}}{4}.$$

Also $L \ge n_b$, while $j \le R \le \sqrt{n} \le n/2$ and $n \ge 2T$ imply $n - j \ge n/2 \ge T$. Then, for $L \le j \le R$,

$$0 \le \sqrt{n} - \sqrt{n-j} = \frac{j}{\sqrt{n} + \sqrt{n-j}} \le \frac{j}{\sqrt{n}} \le 1.$$

Because $0 < 1 - j/n \le 1$ and $0 \le \sqrt{n} - \sqrt{n-j} \le 1$, the ratio $V(n-j)/V(n)$ satisfies

$$\frac{V(n-j)}{V(n)} = \left(1 - \frac{j}{n}\right)^{-\gamma} \exp\left(B(\sqrt{n-j} - \sqrt{n})\right) \ge e^{-B}.$$

For $L \le j \le R$, there are at least $\sqrt{n}/4$ terms, so their total contribution is at least

$$\frac{\sqrt{n}}{4} 0.01 c_U e^{-B}\, V(n). \tag{6.21}$$

Since $4 \times 1.99/0.01 = 796$, the lower bound (6.21) strictly exceeds (6.20) when $\sqrt{n} \geq K(B,\gamma)N_b + 1$. Thus the convolution is positive for every $n \geq N_b^*$, proving the claim.

Denote

$$N_0(r,s,k) := \delta_0 + N_b^*.$$

Taking coefficients in (6.17) gives

$$H(n) = \sum_{j=0}^{n-\delta_0} b(j)u(n-\delta_0-j).$$

The claim therefore gives $H(n) > 0$ for every $n \geq N_0(r,s,k)$ when $r \geq 4$.

Finally, for arbitrary R, S in the stated principal range, let

$$d = (R,S), \qquad r = R/d, \qquad s = S/d,$$

and define

$$N(R,S,k) := \begin{cases} 0, & r \in \{2,3\}, \\ dN_0(r,s,k), & r \geq 4. \end{cases} \tag{6.22}$$

The cases $r = 2$ and $r = 3$ are Corollary 4.5 and [12, Theorem 1.7], respectively. Since the series for (R,S) is obtained from the one for (r,s) by replacing q with q^d, this proves Theorem 1.4. The corresponding Notebook commands are

`GeneralThreshold[R,S,k]`, `MercaReport[MercaSystem[r,s,k]]`

in `Merca_Original_Notebook_Updated.nb`.

□

In summary, we state the systematic method to prove Merca's conjecture 1.3 by determining the constant $N(r,s,k)$ for given r, s and k such that when $n \geq N(r,s,k)$ the conjecture holds.

7 Concluding remarks

For the partition functions considered in this paper, we can consider the following more general form

$$(-1)^k \frac{\sum_{j=k}^{\infty}(-1)^j q^{rj(j+1)/2}\left(q^{-sj} - q^{(j+1)s}\right)}{(q^s;q^r)_{\ell_1}(q^{r-s};q^r)_{\ell_2}}, \tag{7.1}$$

where r and s are coprime positive integers with $1 \leq s < r/2$, $k \geq 1$, $\ell_1 \geq 0$, $\ell_2 \geq 0$. By choosing other values of ℓ_1 and ℓ_2, one may investigate other truncated theta series. Moreover, following our method, one may show the nonnegativity of the coefficients of the following truncated quintuple product series,

$$\frac{\sum_{j=-k}^{k}(-1)^j q^{Rj(3j+1)}\left(q^{3jS} - q^{-(3j+1)S}\right)}{(q^{-S},q^{R+S},q^R;q^R)_\infty (q^{R-2S},q^{R+2S};q^{2R})_\infty},$$

which was studied by Chan, Ho and Mao in [8]. Finally, we propose the following extension of Conjecture 1.1.

Conjecture 7.1. *For positive integers $1 \leq S < R/2$, $\ell \geq 1$, $k \geq 4$ and $k > \ell$, there exists a constant N depending on R, S and k such that the truncated theta series*

$$\frac{(-1)^{k-1}\left(\sum_{j=-k+1}^{k} - \sum_{j=-\ell+1}^{\ell}\right)(-1)^j q^{Rj(j+1)/2+Sj}}{(q^S,q^{R-S},q^R;q^R)_\infty} \tag{7.2}$$

has nonnegative coefficients when $n \geq N$.

Obviously, if $k-\ell \equiv 1 \pmod 2$, the sums in the numerator of the above series can be written as $(-1)^{k-1}\sum_{j=-k+1}^{k}+(-1)^{\ell-1}\sum_{j=-\ell+1}^{\ell}$. Then it is just a direct corollary of Conjecture A-M-conj and thereby the conjecture holds for $N=0$. Thus we obtain the following result on partition functions.

Theorem 7.2. *When $k-\ell \equiv 1 \pmod 2$, we have*

$$(-1)^{\ell}\left(\sum_{j=-k+1}^{-\ell}+\sum_{j=\ell+1}^{k}\right)(-1)^{j}P_{R,S}(n-(Rj(j+1)/2+Sj))\geq 0,$$

where $P_{R,S}(n)$ denotes the number of partitions of n with parts congruent to $\pm S$ modulo R.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (Grant No. 12071235), and the Fundamental Research Funds for the Central Universities.

Declarations

Competing interests: The authors declare no competing interests.

Availability of data and materials: Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Code availability: The Mathematica file used to reproduce the finite coefficient evaluations, the final parameter-dependent threshold, and the compact-interval bounds is supplied as supplementary material together with an English user guide.

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